Kidney & Upper-Tract Tumours Rapid Revision

Campbell-Walsh-Wein · Chapters 136–139, 141–144

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 5h 40min even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 141* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
136	Benign Renal Tumours	21 min
137	Malignant Renal Tumours	71 min
138	Urothelial Tumours of the Upper Tract & Ureter	28 min
139	Surgical Management of UTUC	46 min
141	Contemporary Open Surgery of the Kidney	54 min
142	Laparoscopic & Robotic Surgery of the Kidney (incl. donor nephrectomy)	46 min
143	Focal Therapy for Renal Tumours	28 min
144	Treatment of Advanced RCC	36 min
★	Putting it together: the big treatment choices	10 min
	Total (23825 words, at 70 words/min)	**5h 40min**

136 Benign Renal Tumours

Reading time ≈ 21 min (slow pace)

1 · THE BENIGN SMALL RENAL MASS

- Up to **~25%** of surgically treated small renal masses (SRMs) are benign.
- Ultrasonography (US), computed tomography (CT) and magnetic resonance imaging (MRI) cannot reliably separate benign from renal cell carcinoma (RCC). Only the **simple cyst** and **fat-containing angiomyolipoma (AML)** are diagnosable on imaging. Small size, female sex and older age are unreliable pointers.
- Management follows RCC risk: no follow-up (simple cyst) → selective arterial embolisation (SAE) (large AML) → excision (solid or complex cystic mass). Renal mass biopsy and the SRM algorithm → *Ch 137*.

2 · RENAL CYSTS AND THE BOSNIAK CLASSIFICATION

- Commonest renal mass (↑ with age, male sex, renal dysfunction); usually asymptomatic; grow 1–2 mm/yr.
- **Simple cyst**: any size, homogeneous, **≤20 Hounsfield units (HU)** → no further imaging. Heterogeneity, >20 HU, septa, wall thickening, nodules or calcification → Bosniak staging.
- **Version 2019** added MRI criteria, mainly because too many resected Bosniak III lesions proved benign.

Class	Features (v2019)	Malignancy	Management (Canadian Urological Association, CUA 2023)
I	Thin (**≤2 mm**) smooth wall (may enhance); simple fluid; no septa or calcification	Benign	No follow-up
II	Thin wall plus (a) **≤3** thin enhancing septa, any calcification; or (b) homogeneous marked T2 hyperintensity; or (c) marked T1 hyperintensity	Highly likely benign	No follow-up
IIF	Minimally thickened (**3 mm**) enhancing wall/septum, or **≥4** thin septa; or heterogeneous T1 hyperintensity on fat-saturated MRI	Likely benign	Active surveillance up to **5 yr**
III	Enhancing thick (**≥4 mm**) or irregular (≤3 mm obtuse-margined protrusion) wall/septum	**~50%**	≤2 cm: surveillance preferred; 2–4 cm: surveillance or excision (shared decision); >4 cm: excision, partial (PN) over radical nephrectomy (RN) when feasible
IV	Enhancing nodule: ≥4 mm obtuse-margined, or any-size acute-margined protrusion	**~90%**	

KEY POINT Memory ladder: wall ≤2 → 3 → ≥4 mm → irregularity → nodule. Traps: any calcification is compatible with II; a Bosniak I wall may enhance; large (≥3 cm) hyperdense cysts are no longer IIF. Bosniak I/II growth is expected and not a sign of malignancy.

KEY POINT **Why surveil small III/IV?** Bosniak III has a low positive predictive value, and cystic RCC is indolent: much of its volume is fluid, so tumour burden is low and size predicts outcome poorly (proposed stage **pT1c**). The number needed to treat to prevent one metastasis is therefore very high.

- European Association of Urology (EAU): manage Bosniak IV as a localised solid mass. Biopsy only if a significant solid component and the result would change management. Thermal ablation for cystic masses: effectiveness uncertain.

Diffuse cystic disease

	Autosomal dominant polycystic kidney disease (ADPKD)	Acquired cystic kidney disease (ACKD)
Cause	PKD1 (chr 16, polycystin-1); PKD2 (chr 4, polycystin-2) — ciliary calcium signalling	End-stage renal disease (ESRD) on dialysis; cysts from proximal tubule; **80%** by 10 yr
Kidneys	Grossly enlarged; a cause of ESRD	Small/normal; a consequence of ESRD
RCC	Risk not raised, but earlier, more sarcomatoid, harder to diagnose	Risk **≥10×**; multifocal, bilateral, male; papillary commoner; distinct World Health Organization (WHO) 2022 entity

- ADPKD: hypertension (renin–aldosterone, ischaemia); extrarenal cysts (liver, pancreas, spleen, ovary, seminal vesicle, arachnoid). US diagnosis in PKD1 relatives aged 15–29: **≥3 cysts**; a normal scan excludes disease reliably only from age 40.
- Nephrological care: blood pressure control, lifestyle; **tolvaptan** (vasopressin V2 antagonist) slows decline in rapid progressors.
- Refer to urology for stones, flank pain, gross haematuria, cyst haemorrhage (self-limiting in days) or cyst infection (fever, raised inflammatory markers; confirm with fluorodeoxyglucose positron emission tomography (FDG-PET); empirical fluoroquinolone or trimethoprim–sulfamethoxazole).
- Intervention for bulk pain only: aspiration, embolisation, fenestration; nephrectomy in selected patients only (morbidity). ESRD → transplantation.

3 · ONCOCYTOMA

- **Commonest benign enhancing renal mass**; up to half of benign SRMs <4 cm.
- Syndromic in **Birt–Hogg–Dubé** (autosomal dominant, folliculin, chr 17): facial fibrofolliculomas, lung cysts with pneumothorax, oncocytoma and chromophobe RCC (chRCC). Renal oncocytosis (diffuse bilateral nodules → chronic kidney disease, CKD) may harbour chRCC/hybrid tumours.
- Imaging: central **stellate scar** (non-specific); lower corticomedullary attenuation, gradual enhancement and **segmental enhancement inversion** favour oncocytoma.

	Oncocytoma	Chromophobe RCC
Cytokeratin 7 (CK7)	Rare, single-cell; KIT positive	Diffuse positive
Cytogenetics	Normal, −1, −Y, 11q13 rearrangement	Extensive abnormalities

> **KEY POINT** **Why can a core not clear oncocytoma?** Sampling error, and chRCC contains oncocytoma-like areas — a substantial minority of biopsy "oncocytomas" are RCC at surgery. EAU: definitive diagnosis is not possible on a needle core; adequate tissue with CK7 and cytogenetics helps.

4 · ANGIOMYOLIPOMA

Pathology and genetics

- Perivascular epithelioid cell tumour: dysmorphic vessels, smooth muscle, fat. Sporadic (women, 4th–5th decade) or tuberous sclerosis complex (TSC; AML in **55–90%**) or lymphangioleiomyomatosis (LAM).
- Biallelic loss of **TSC2** (tuberin, 16p13) or TSC1 (hamartin, 9q34). Hamartin–tuberin normally inhibit mammalian target of rapamycin (mTOR); loss → unchecked growth and angiogenesis. Multiple tumours are clonally independent.
- Immunohistochemistry: **HMB-45, Melan-A**, smooth muscle actin positive; keratin negative.

	Sporadic	TSC-related	Epithelioid
Lesions	Solitary, smaller	Bilateral, multifocal, recurrent; haemorrhage, hypertension, CKD	Minimal fat; aggressive (necrosis, venous extension, metastasis)

	Sporadic	TSC-related	Epithelioid
Surveillance	US/CT at 6 and 12 months, then yearly	MRI every 1–3 yr	—
Treatment	If triggers met (below)	Everolimus if **>3 cm**	As RCC

Diagnosis

- Macroscopic fat (**−15 to −20 HU** on non-contrast CT) is diagnostic. US unreliable (some RCCs hyperechoic). MRI: signal loss on fat suppression, India-ink artefact. Fat-poor AML mimics RCC → MRI/biopsy.
- Giant AML arises from parenchyma; retroperitoneal liposarcoma envelops and compresses the kidney.

Management

ALGORITHM

Spontaneous retroperitoneal haemorrhage (Wunderlich†) → resuscitate → **SAE** → nephrectomy only if SAE unavailable or unsuitable

Symptomatic (pain, haematuria, barriers to care) → SAE → nephron-sparing surgery (NSS) or RN → ablation in selected cases

Asymptomatic sporadic: >**6 cm**, aneurysm **≥5 mm**, growth >**0.25 cm/yr**, childbearing age/pregnancy → treat; otherwise surveil

TSC: >3 cm or growing → everolimus first line; SAE or kidney-sparing resection second

> **KEY POINT** **Why abandon 4 cm?** Bleeding risk rises steeply with size but the 4 cm cut-off was poorly specific, so most size-treated patients were overtreated. AML <4 cm → surveillance; observed AML ≥4 cm → thin-slice CT angiography to detect aneurysm. "Barriers to care" is an indication because surveillance presumes access to imaging and emergency care. Pregnancy raises bleeding risk and stakes → consider prophylactic treatment before conception.

- **SAE**: first line for acute bleeding, symptoms, prophylaxis and pre-surgery. Post-embolisation syndrome (fever, pain, leucocytosis, vomiting) is common and self-limiting; abscess and renal infarction occur; major complications rare.
- **NSS**: preferred operation — preserves function, crucial in TSC because lesions keep recurring.
- TSC consensus: haemorrhage → embolise then corticosteroids; avoid nephrectomy. EAU (sporadic): mTOR inhibitor only if unsuitable for SAE/surgery.

5 · OTHER BENIGN EPITHELIAL AND MESENCHYMAL TUMOURS

- **Papillary adenoma**: unencapsulated, **≤15 mm** (≤5 mm until 2015); shares +7, +17, −Y and KMT2C with papillary RCC; no metastatic capacity; acceptable in a donor kidney.
- **Metanephric adenoma**: women, 50s; may cause polycythaemia. CT: hypovascular, mild homogeneous enhancement — not diagnostic, and biopsy accuracy unproven → excise by NSS. **Wilms tumour 1 (WT1), CD57 positive; CK7, AMACR negative**; lacks papillary RCC's +7/+17/−Y.
- **Mixed epithelial and stromal tumour (MEST) and adult cystic nephroma**: one WHO 2022 family; middle-aged women; oestrogen/progesterone-receptor-positive stroma; multiseptated, mimic cystic RCC → manage by Bosniak guidelines, consider biopsy; rare malignant change. Paediatric cystic nephroma is distinct (**DICER1**).
- **Leiomyoma**: from renal capsule; T1/T2 hypointense; smooth muscle actin, desmin, caldesmon positive. Often actually lipid-poor AML: both may be HMB-45 positive, but **Melan-A** is positive only in AML. NSS if surgery.

Tumour	Key trait
Haemangioma	Sturge–Weber, Klippel–Trénaunay; usually excised
Haemangioblastoma	Von Hippel–Lindau (VHL); mimics RCC; cured by excision
Lymphangioma	Perinephric cysts; chyluria; aspiration/sclerosis or excision
Juxtaglomerular cell tumour (reninoma)	Young women; refractory hypertension, hypokalaemia, high renin; ≤3 cm; renin granules pathognomonic; excision cures most
Solitary fibrous tumour	~10% aggressive: age ≥55, ≥15 cm, mitoses, necrosis

6 · LANDMARK TRIAL

Trial	Design	Result	Take-home
EXIST-2	Phase III randomised, everolimus vs placebo, TSC/LAM AML	≥50% volume reduction **42% vs 0%**; no bleeding requiring embolisation	Approved; mTOR inhibition first line in TSC

7 · MCQ ONE-LINERS

1. Simple cyst: ≤20 HU.
2. Bosniak III ~50%, IV ~90% malignant.
3. ≤3 thin septa = II; ≥4 = IIF.
4. IIF: surveillance up to 5 years.
5. Proposed cystic RCC stage: pT1c.
6. PKD1 chr 16; PKD2 chr 4.
7. ACKD cysts arise from proximal tubule.
8. Everolimus: stomatitis, acne, proteinuria, hypercholesterolaemia.
9. Commonest benign enhancing SRM: oncocytoma.
0. Birt–Hogg–Dubé: folliculin, chr 17.
1. Oncocytoma: KIT positive, CK7 single-cell.
2. Segmental enhancement inversion favours oncocytoma.
13. AML: HMB-45, Melan-A positive; keratin negative.
14. TSC2 tuberin 16p13; TSC1 hamartin 9q34.
15. AML intralesional aneurysm ≥5 mm predicts bleeding.
16. TSC AML >3 cm: everolimus.
17. Papillary adenoma ≤15 mm.
18. Metanephric adenoma: WT1, CD57 positive; polycythaemia.
19. DICER1: paediatric cystic nephroma.
20. Leiomyoma: commonest origin renal capsule.
21. Renal haemangioblastoma: VHL.
22. Reninoma: hypertension plus hypokalaemia in young woman.

137 Malignant Renal Tumours

Reading time ≈ 71 min (slow pace)

1 · OVERVIEW, EPIDEMIOLOGY & AETIOLOGY

- Renal cell carcinoma (RCC) = 2–3% of adult cancers; the most lethal common urological cancer. Male:female **2:1**; peak 55–75 y; **4–6% familial**.
- Incidence ↑ ~3%/y, driven by incidental detection on ultrasound (USG) and computed tomography (CT); >60% now incidental, ~70% organ-confined. Because imaging finds many small indolent tumours, incidence has risen far faster than mortality — the basis for today's emphasis on avoiding overtreatment.
- Risk factors: **tobacco** (most accepted; relative risk ~1.4, dose-dependent, falls after quitting), **obesity** ("obesity paradox": higher body mass index but lower stage), **hypertension** (the disease, not its drugs); chronic kidney disease (CKD)/end-stage renal disease (ESRD); family history 2–4×.
- Cell of origin: clear cell & papillary → **proximal tubule**; chromophobe & collecting duct → distal nephron.
- Children/young adults: RCC 1–6% of paediatric renal tumours (Wilms ~90%), equal to Wilms in the 2nd decade; commonest = **Xp11/TFE3 translocation RCC**. They respond well to surgery even when node-positive → formal lymph node dissection (LND) at radical nephrectomy (RN) recommended.

2 · CLASSIFICATION & PATHOLOGY

World Health Organization (WHO) 2022 classification (key groups)

Group	Entities
Renal cell tumours	Clear cell; multilocular cystic neoplasm of low malignant potential; papillary; chromophobe; other oncocytic; RCC not otherwise specified
Molecularly defined (new)	TFE3/TFEB-rearranged; ELOC-mutated; fumarate hydratase (FH)-deficient; succinate dehydrogenase (SDH)-deficient; ALK-rearranged; SMARCB1-deficient medullary
Other	Collecting duct; mucinous tubular & spindle cell; tubulocystic; acquired cystic disease-associated; eosinophilic solid & cystic; clear cell papillary tumour
Non-RCC malignant	Urothelial/squamous; sarcomas; nephroblastic; neuroendocrine; lymphoma; **epithelioid angiomyolipoma**; metastases

> **KEY POINT** Trap: epithelioid angiomyolipoma is malignant, whereas classic angiomyolipoma (AML), papillary adenoma and oncocytoma are benign. The papillary type 1/2 split is **abandoned** — "type 2" proved to be several diseases (MET-altered, MiT translocation, FH-deficient) with very different prognoses, which is why WHO moved to molecular definitions.

Gross features

- Round, usually 4–8 cm, with a **pseudocapsule** (compressed parenchyma and fibrous tissue, not a true capsule); unlike upper-tract urothelial carcinoma, not infiltrative except collecting duct and sarcomatoid. Yellow-tan with necrosis/haemorrhage; cystic degeneration 10–25% (better prognosis); calcification ~10%.
- Aggressive local behaviour in ~20% (perinephric fat, sinus, collecting system); Gerota's fascia usually blocks adjacent-organ spread.
- Venous involvement ~10% (more than any tumour), up to the right atrium. Bilateral 2–4% sporadic; multicentric 10–20%. Small satellite lesions, invisible on imaging, are the main cause of ipsilateral recurrence after partial nephrectomy (PN).

Subtypes

Subtype (share)	Genetics	Hallmarks	Behaviour
Clear cell (70–80%)	3p loss/VHL (>90% 3p altered); +5q; PBRM1, SETD2, BAP1	Golden-yellow, hypervascular; clear cytoplasm (dissolved glycogen, lipid); carbonic anhydrase IX (CA-IX)+; sarcomatoid 5–8%	Most venous invasion; worse localised, better metastatic
Papillary (10–15%)	**Trisomy 7, 17**, loss Y; MET altered 81%	Hypovascular; papillae on fibrovascular cores; CK7, AMACR; multicentric up to 40%	Good (5-y cancer-specific survival 86–92%); common in acquired cystic disease
Chromophobe (3–5%)	Losses 1, 2, 6, 10, 13, 17; TP53, PTEN; FLCN in BHD	"Plant cells", perinuclear halo, raisinoid nuclei; diffuse CK7	Intercalated cells; >90% cancer-free at 5 y unless sarcomatoid
Clear cell papillary tumour (<5%)	—	Linear papillae; CK7; **cup-like CA-IX**	ESRD, VHL; now considered benign
Collecting duct (Bellini)	Multiple losses	Central, infiltrative, desmoplastic; PAX8+; diagnosis of exclusion	Poor; urothelial-like → may respond to cisplatin/gemcitabine
SMARCB1-deficient medullary	**Sickle cell trait**	Young Black patients; infiltrative; SMARCB1 loss	Dismal, median 8–13 mo; modest platinum response
FH-deficient	FH (1q42–43)	Macronucleolus with perinucleolar halo; FH loss, 2SC+	Metastasises early
MiT translocation	Xp11.2 TFE3, TFEB	Clear cells + papillae; fluorescence in situ hybridisation (FISH) confirms	40% of paediatric RCC; t(X;1) late nodal recurrence
SDH-deficient	SDH genes	Vacuolated eosinophilic cells; SDHB loss	Young; usually favourable; often germline
ALK-rearranged	ALK (2p23)	Test ALK before labelling "unclassified"	Dramatic responses to ALK inhibitors in some
Unclassified (1–5%)	—	Catch-all, shrinking	Poor

- **Indolent:** multilocular cystic, tubulocystic, mucinous tubular & spindle cell, eosinophilic solid & cystic, ELOC, clear cell papillary.
- **Lethal:** medullary, FH-deficient, collecting duct, unclassified, sarcomatoid.
- Chromophobe vs oncocytoma is hard even for experts → WHO 2022 "oncocytic renal neoplasm of low malignant potential" (indolent). Papillary adenoma ≤15 mm is benign.
- **Sarcomatoid** (1–5%): a dedifferentiated area of any subtype, not a separate tumour; median survival <1 y. Sarcomatoid or rhabdoid change = automatic grade 4 and must be reported in every RCC.

> **KEY POINT** Why does localised clear cell RCC do worse than papillary/chromophobe, yet metastatic clear cell does better? It is intrinsically more aggressive, but its VHL-driven angiogenesis and immunogenicity make it respond to tyrosine kinase inhibitors (TKIs) and immune checkpoint inhibitors (ICIs).

WHO/International Society of Urological Pathology (ISUP) grading — replaced Fuhrman; validated for clear cell & papillary

Grade	Criterion
1	Nucleoli absent/inconspicuous, basophilic at ×400
2	Nucleoli conspicuous, eosinophilic at ×400; not prominent at ×100
3	Nucleoli conspicuous, eosinophilic at **×100**
4	Extreme pleomorphism, giant cells, **rhabdoid/sarcomatoid**

Grades 1–3 turn on the magnification at which nucleoli become visible; grade 4 is morphological. Nuclear grade is an independent prognostic factor.

3 · MOLECULAR BIOLOGY & HEREDITARY SYNDROMES

Von Hippel-Lindau (VHL)–hypoxia-inducible factor (HIF)–vascular endothelial growth factor (VEGF) pathway

ALGORITHM

Normoxia: VHL protein + elongin B/C + cullin-2 + RBX1 = E3 ubiquitin ligase → HIF-1α/2α degraded

VHL loss or hypoxia → **HIF-2α** accumulates → dimerises with ARNT → ↑ VEGF (angiogenesis), PDGF, TGF-α, GLUT1, erythropoietin, CA-IX → hypervascular clear cell RCC

Drug targets: VEGF ligand (bevacizumab) · VEGF receptor TKIs (sunitinib, pazopanib, axitinib, cabozantinib) · mTOR (temsirolimus, everolimus) · HIF-2α (**belzutifan**, US Food and Drug Administration (FDA) 2021 for VHL, ~50% response) · CTLA-4 (ipilimumab), PD-1 (nivolumab, pembrolizumab), PD-L1 (atezolizumab, avelumab).

- Biology explains therapeutic history: multidrug-resistance pumps → chemotherapy fails (response 5–7%); VHL-driven vascularity → anti-angiogenics work; immune evasion via PD-L1/PD-1 → checkpoint blockade is now the mainstay → *Ch 144*.
- Knudson two-hit: familial patients inherit one hit in every cell → early, bilateral, multifocal tumours. All familial RCC syndromes are **autosomal dominant**; all are tumour-suppressor losses except **MET (oncogene)**.
- Checkpoints: T-cell activation balances positive signals (T-cell receptor–antigen; B7–CD28) against negative ones (B7–CTLA-4; tumour PD-L1–T-cell PD-1). Blocking the negative side tilts towards tumour killing. Immune adverse events: colitis, hepatitis, dermatitis, pneumonitis.
- Poor prognosis in all subtypes: **CDKN2A** (9p) loss, CpG island hypermethylation, Th2 immune signature. BAP1 mutation (up to 14% sporadic clear cell) → necrosis, high grade, rapid growth, unsuitable for active surveillance (AS); test germline BAP1 in early-onset multifocal VHL-negative clear cell RCC.
- c-MET is upregulated in most subtypes; cabozantinib hits VEGF receptor, MET and AXL, explaining second-line activity.

Hereditary RCC syndromes

Syndrome	Gene (chr)	Renal tumour	Other features	Renal management
VHL	VHL (3p25)	Clear cell, bilateral, multifocal (~50%)	central nervous system (CNS)/retinal haemangioblastoma, phaeochromocytoma, pancreatic & renal cysts, endolymphatic sac tumour, epididymal cystadenoma	AS <3 cm; nephron-sparing ≥3 cm; belzutifan
Hereditary papillary RCC	MET (7q31) oncogene	Multiple bilateral papillary	None	AS <3 cm; nephron-sparing ≥3 cm

Syndrome	Gene (chr)	Renal tumour	Other features	Renal management
Hereditary leiomyomatosis & RCC (HLRCC)	FH (1q42–43)	Aggressive, often **solitary**; collecting duct	Cutaneous & uterine leiomyomas; uterine leiomyosarcoma	**Prompt excision at any size, wide margins**; PN only if wide margin feasible
SDH-RCC	SDHA/B/C/D, SDHAF2	Variable, onset ≤40 y	Paraganglioma/phaeo, gastrointestinal stromal tumour (GIST), thyroid	Excision, PN preferred
Birt-Hogg-Dubé (BHD)	FLCN (17p11.2)	Chromophobe, hybrid oncocytic, oncocytoma	Fibrofolliculomas, lung cysts, pneumothorax	AS <3 cm; nephron-sparing ≥3 cm
Cowden	PTEN (10q23)	Mostly papillary (lifetime 34%)	Breast (50%), thyroid; KILLIN if PTEN-negative	Screen CT/MRI; 3-cm rule
Tuberous sclerosis	TSC1 (9q34)/TSC2 (16p13)	Multiple AMLs; clear cell 1–3%	Cardiac rhabdomyoma, angiofibromas, lymphangioleiomyomatosis, epilepsy	AML: surveil <3, everolimus 3–5, embolise/excise ≥5 cm
BAP1	BAP1 (3p21)	High-grade clear cell	Uveal/cutaneous melanoma, mesothelioma	Excision, nephron-sparing
MiTF-associated	MiTF p.E318K	RCC (>5×; activates HIF-1α)	Melanoma	Undefined

- **VHL genotype–phenotype:** type 1 (deletion/truncation) = no phaeochromocytoma; type 2 (missense) = phaeochromocytoma — 2A low RCC, 2B high RCC, 2C phaeochromocytoma only.
- VHL: RCC in ~50%, now the commonest cause of death. Haemangioblastoma is the commonest lesion; retinal angioma/phaeochromocytoma appear earliest, RCC latest (~39 y) — so screening starts in childhood, renal imaging in adolescence.
- Suspect VHL with early-onset/multifocal clear cell RCC, or RCC plus visual/neurological history, pancreatic cysts, epididymal or inner-ear lesions, or family history of blindness/CNS tumours.
- HLRCC: renal penetrance 15–20%, but nearly all have skin leiomyomas/fibroids (often prior hysterectomy). fluorodeoxyglucose positron emission tomography (FDG-PET) detects small metastases.
- Hereditary papillary: indolent → avoid overtreatment (CKD from repeated surgery). TSC: hamartin/tuberin restrain mTOR → everolimus shrinks AMLs.

4 · CLINICAL PRESENTATION

- **Classic triad** (flank pain, gross haematuria, mass) is rare and signals advanced disease ("too late triad").
- Advanced-disease clues: weight loss, bone pain, cervical nodes, **non-reducing or right-sided varicocele**, leg oedema (inferior vena cava [IVC] obstruction).
- Spontaneous perirenal haematoma without cause: >50% harbour a tumour (AML or RCC) → repeat CT after a few months.

Paraneoplastic syndromes (10–20%)

The kidney normally makes renin, erythropoietin, vitamin D and prostaglandins; RCC overproduces these plus parathyroid hormone-related peptide (PTHrP) and cytokines.

Syndrome	Frequency / mechanism	Key point
Raised erythrocyte sedimentation rate (ESR)	Commonest	Advanced disease marker
Hypercalcaemia	Up to 13%; PTHrP or osteolytic metastases	**Only one manageable medically**: hydration → furosemide; zoledronic acid 4 mg IV 4-weekly (withhold in renal insufficiency); denosumab if refractory; steroids, calcitonin. Definitive: nephrectomy ± metastasectomy
Hypertension	Renin, artery compression, arteriovenous fistula	—
Polycythaemia	Erythropoietin	—
Stauffer syndrome	3–20%; interleukin-6	Non-metastatic liver dysfunction (↑ alkaline phosphatase almost always); exclude liver metastases; normalises after nephrectomy in 60–70%; persistence = viable tumour, poor prognosis
Others	Rare	Cushing, hyperglycaemia, galactorrhoea, neuromyopathy, clotting disorders, cerebellar ataxia

- Hypercalcaemia is non-specific (nausea, anorexia, fatigue, reduced reflexes). From extensive osteolytic metastases it is harder to palliate — bisphosphonates and focused radiotherapy.
- Rule: paraneoplastic syndromes resolve only with excision or systemic therapy (except medically managed hypercalcaemia); persistence after nephrectomy implies residual disease.

5 · DIAGNOSIS

Risk of malignancy

- Strongest predictors: **male sex (~3×)** and **size**. Benign: 20–30% of ≤4 cm, ~40% of ≤2 cm; at ≥7 cm few benign, half high grade. Clinical nomograms are little better than a coin toss (concordance 0.55–0.65).
- Clinical T1 mass: ~20% benign (oncocytoma, fat-poor AML) · ~60% indolent RCC · ~20% potentially aggressive.

Imaging

- **Multiphase CT** (without + with contrast) = single most important test: enhancement **>15–20 Hounsfield units (HU)** suggests RCC; fat **<−20 HU** = AML. Magnetic resonance imaging (MRI): enhancement >20%; best for IVC thrombus and adjacent-organ invasion; macrocyclic gadolinium is safe even on dialysis (nephrogenic systemic fibrosis now near-absent).
- USG simple cyst: smooth wall, round, anechoic, posterior enhancement — anything else needs cross-sectional imaging. Contrast-enhanced USG (microbubbles) if CT/MRI contraindicated — reasonable, not first choice.
- Contrast safety: eGFR <45 receiving iodinated contrast → periprocedural hydration; avoid where possible in severe CKD.
- FDG-PET: limited (renal background uptake), not for staging. **Sestamibi single-photon emission CT (SPECT)/CT**: mitochondria-rich **oncocytoma**/hybrid tumours take it up. ^{124}I-girentuximab PET targets CA-IX (clear cell).
- Enhancement: clear cell & oncocytoma strong (overlap — cannot separate); papillary/chromophobe moderate. Infiltrative pattern → lymphoma, urothelial carcinoma, sarcomatoid, collecting duct, medullary, xanthogranulomatous pyelonephritis.
- Complexity scores — **R.E.N.A.L.** (Radius, Endophytic/exophytic, Nearness to collecting system, Anterior/posterior, Location to polar lines), PADUA, C-index — guide PN vs RN, approach and complication risk.

ALGORITHM

Simple cyst → observe

Complex/solid: estimated glomerular filtration rate (eGFR) >45 → renal CT or MRI with/without contrast · eGFR 30–45 → hydrate, same · eGFR <30 → non-contrast CT or gadolinium MRI · contrast allergy → USG/contrast USG or MRI

Fat → AML · complex cyst → Bosniak · solid enhancing → management options ± renal mass biopsy

Bosniak

Full v2019 table & management → *Ch 136*. Malignancy: IIF 3–10% · III ~50% · IV 75–90%.

- Key discriminator: unequivocal **measurable enhancement** (absent in II/IIF; present in III; enhancing soft tissue in IV).
- Avoid thermal ablation and biopsy in cystic masses — poor yield and risk of tumour spillage. Cystic RCC is indolent, so surgery for III/IV is no longer mandatory; AS is an option.

Renal mass biopsy (RMB)

- Utility-based: obtain when the result may change management, and when lymphoma, metastasis, infection or inflammation is suspected. Not needed in young fit patients heading for surgery regardless, or frail patients managed conservatively regardless. **Multiple cores preferred over fine-needle aspiration**.
- Safe: haematoma ~5%, transfusion <0.5%, seeding almost unreported. Specificity 94%, positive predictive value 98.8%; **non-diagnostic ~14%** (reduced by repeat); subtype accuracy ≥90%, grade 60–80%. A non-malignant result does not prove benignity.
- Downsides: non-diagnostic results, occasional false negatives, spillage in cystic tumours. TA is less successful in avidly enhancing/clear cell tumours, so biopsy before ablation. Molecular profiling of biopsies is a research priority.

Screening

- No population screening (low yield; most small masses indolent). Not in autosomal dominant polycystic kidney disease.
- ESRD (acquired cystic disease in 80%; RCC in 1–2% of these): USG/CT from 3rd year of dialysis if fit. Continue native-kidney imaging after transplant.
- VHL protocol: eye/physical exam yearly from infancy; metanephrines from 5; CNS MRI + audiogram 2-yearly from 11; **abdominal MRI 2-yearly from 15**.
- Genetic counselling: RCC at **≤46 y**, multifocal/bilateral, suggestive personal/family history or pathology.

6 · STAGING & PROGNOSIS

tumour–node–metastasis (TNM), American Joint Committee on Cancer 8th	Definition	5-y survival
T1a / T1b	≤4 / >4–7 cm, confined	95–100 / 85–95%
T2a / T2b	>7–10 / >10 cm	65–85 / 55–75%
T3a	Renal vein/segmental branches, **pelvicalyceal system**, perirenal/sinus fat, within Gerota's fascia	45–75%
T3b	IVC below diaphragm	40–55%
T3c	IVC above diaphragm or **caval wall invasion**	20–45%
T4	Beyond Gerota's, incl. **contiguous ipsilateral adrenal**	0–35%

tumour–node–metastasis (TNM), American Joint Committee on Cancer 8th	Definition	5-y survival
N1 / M1	Regional node / distant (non-contiguous adrenal = M1)	0–30%

Stage I T1N0 · II T2N0 · III T1–2N1 or T3 any N · IV T4 or M1. Edition changes: 6th split T1 at 4 cm; 7th split T2 at 10 cm, contiguous adrenal → T4, isolated renal vein thrombus T3b → T3a, N1/N2 merged; 8th pelvicalyceal invasion → T3a.

> **KEY POINT** TNM limitation: stage is set by the single worst feature, so co-existing adverse features (e.g. adrenal invasion plus supradiaphragmatic thrombus) are lost — record all, and use integrated nomograms.

- **Staging work-up:** abdominal CT + chest X-ray (chest CT if symptoms, abnormal X-ray or high risk); bone scan only with ↑ alkaline phosphatase, bone pain or poor performance status; no PET. Nodes ≥2 cm almost always malignant; smaller nodes often reactive and should not preclude surgery. CT sensitivity: renal vein thrombus 78%, IVC 96% (misses are right-sided). Abnormal/indistinct adrenal → treat as involved; normal → leave it.
- **Pathological stage** = single most important factor; then grade, size, subtype, necrosis, sarcomatoid. Big drop with T4, N1, M1 (node-positive 5-y 5–30%).
- Venous thrombus with otherwise confined disease: 5-y 45–69% — high stage, still curable; worse with nodes/metastases. Venous wall invasion is as important as level. Renal sinus fat (access to sinus veins) may be riskier than perinephric fat; collecting-system invasion independently adverse.
- Adverse clinical factors: symptoms, weight loss >10%, poor performance status, anaemia, hypercalcaemia, ↑ ESR/lactate dehydrogenase (LDH)/alkaline phosphatase, thrombocytosis.
- Integrated nomograms (e.g. Kattan) outperform stage but need revalidation. ASSURE post hoc: recurrence algorithms barely beat stage + grade → AUA risk groups use stage/grade. † **UISS** = TNM + grade + Eastern Cooperative Oncology Group performance status; **SSIGN** (Mayo) = Stage, Size, Grade, Necrosis; **Leibovich** = pT, pN, size, grade, necrosis → recurrence risk in clear cell RCC.

7 · MANAGEMENT OF LOCALISED RCC

American Urological Association (AUA) 2021 principles

- **Work-up:** multiphase imaging, metabolic panel, blood count, urinalysis, chest imaging; assign **CKD stage (eGFR + proteinuria)** — both independently predict functional decline and survival. Refer to nephrology if GFR <45, confirmed proteinuria, diabetic CKD, or expected post-op GFR <30.
- **Counsel** on tumour biology (many cT1a are low risk), morbidity of each option, age, comorbidity, frailty, life expectancy and CKD/dialysis risk.
- **PN prioritised for cT1a** when intervention is indicated; nephron-sparing prioritised for solitary kidney, bilateral or familial tumours, CKD/proteinuria; consider in the young, multifocal disease, or comorbidity threatening future function.
- **RN preferred only if all 3**: (1) high complexity, PN hard even for experts; (2) no CKD/proteinuria; (3) normal contralateral kidney with new baseline GFR likely >45. Otherwise PN, unless overriding safety/oncological concerns.
- Thermal ablation (TA): alternative for cT1a **<3 cm**; percutaneous preferred; RMB before (preferred) or at TA.
- LND for clinically positive nodes; adrenalectomy only if imaging/operative suspicion; minimally invasive where oncology is not compromised. Examine adjacent parenchyma for medical renal disease; refer to oncology for metastasis or incomplete resection.

Renal function: why nephron-sparing matters

- ~25% of "ideal" PN candidates already have CKD (GFR <60); CKD after RN 65% vs PN 20%.

- **Surgical CKD** (one-time nephron loss) is stable (~0.7%/y) with near-normal survival if GFR >45; **medical CKD** (hypertension, diabetes) progresses (~4.7%/y) and shortens life. Hence the RN rule hinges on a healthy contralateral kidney and GFR >45.
- Retrospective PN survival advantages reflect selection bias (PN cannot be oncologically superior): cancer-specific survival depends on size/stage, overall survival on age and comorbidity.
- PN preserves ~90% of global function, RN 65–70%; compensatory hypertrophy 20–30%.
- **Predicted GFR after RN ≈ global GFR × contralateral split function × 1.25** (split function by volumetry).

Active surveillance

- Small masses grow 0.1–0.3 cm/y; metastasis <2%. **DISSRM** registry supports AS; delayed intervention usually still allows nephron-sparing.
- Favouring AS: elderly, life expectancy <5 y, comorbid, frail, marginal renal function; tumour <3 cm, growth <5 mm/y, cystic, non-infiltrative, low complexity. Strong consideration for solid <2 cm or predominantly cystic.
- Imaging every 3–6 mo (AS, with intent to treat) or 6–12 mo (expectant management, often palliative). Repeat at 3–6 mo, consider RMB, then by growth rate.
- AS implies intent to intervene on progression; expectant management is less intensive, often palliative. Adjuncts: creatinine, proteinuria, liver function tests, chest imaging; Doppler or diffusion-weighted MRI if contrast contraindicated.
- **Triggers to intervene**: size >3 cm, growth >5 mm/y, stage progression, adverse biopsy, symptoms. Caution: a subgroup grows fast and delay may lose the PN/TA window.

Partial nephrectomy

Indication	Situations
Imperative	Bilateral RCC, solitary functioning kidney — RN would mean dialysis
Relative	Contralateral kidney threatened (renal artery stenosis, stones, diabetes, hypertension, hydronephrosis)
Elective	cT1a with normal contralateral kidney (standard of care)

- **Pre-op:** exclude local extension/metastasis; 3-D contrast CT (MRI in severe CKD) to map vessels and collecting system.
- **Steps:** temporary pedicle occlusion → excision with a rim of normal tissue → collecting-system closure and vessel ligation → capsular reconstruction → *Ch 142*. Negative margin is the priority; its width is immaterial (critical for hilar tumours).
- **Function determinants:** pre-op parenchymal quality (fixed ceiling) → **quantity of vascularised parenchyma preserved (primary, modifiable)** → ischaemia (secondary if limited). Use hypothermia if ischaemia ≥25–30 min, especially solitary kidney/CKD. Off-clamp shows no functional gain over limited warm ischaemia; segmental clamping and 3-D printing are promising.
- **Complications:** urine leak and bleeding commoner than after RN (highly vascular reconstructed organ); most managed conservatively. Nephrometry scores predict them. Laparoscopic/robotic ≈ open in experienced hands.
- Larger (pT1b/T2), complex or hilar tumours: PN safe in selected patients; pursue whenever function matters.
- Elective PN for cT1a: oncology equals RN; local recurrence 1–2%, mostly occult multifocal disease away from the bed — balanced by 1–2% contralateral RCC, where RN would have left a solitary kidney with cancer.
- **Solitary kidney:** temporary dialysis ~3.5%, ESRD ~4.5%; need ≥20–30% of one kidney. Loss of >50% parenchyma → hyperfiltration, proteinuria, focal segmental glomerulosclerosis → angiotensin-converting enzyme (ACE) inhibitor + low-protein diet. Local recurrence 3–5%.
- **Bilateral:** staged; PN side first, then RN if needed. If PN impossible for an imperative indication: RN + dialysis (transplant after cancer-free interval), or neoadjuvant TKI (stop several half-lives

before and after surgery — impairs healing).

- **Enucleation** (blunt dissection along pseudocapsule): less bleeding, maximal parenchyma; established in familial RCC; controversial in sporadic, as 40–50% infiltrate the pseudocapsule. Low-grade tumours, chromophobe and oncocytoma rarely invade the capsule → RMB helps select. Record technique (enucleation, standard, enucleoresection) by the surface–intermediate–base (SIB) protocol.

Radical nephrectomy

- Robson principles: early artery then vein ligation, removal outside Gerota's fascia, adrenalectomy, extended LND. Today perifascial dissection remains essential (~25% of cT1b/T2 invade fat), but **no routine adrenalectomy** (unless imaging enlargement or extensive/locally advanced tumour) and **no routine extended LND** (haematogenous spread, variable drainage, no proven benefit).
- Early arterial ligation is accepted, but collaterals in large tumours may prevent hilar control before mobilisation.
- Open: extended subcostal (preferred), midline, thoracoabdominal for very large upper-pole tumours, flank for elderly/high-risk. Laparoscopic RN standard without gross invasion, major venous involvement or bulky nodes — but its ease has driven RN overuse for small masses → *Ch 141*.

Thermal ablation *→ Ch 143*

- Best <2.5–3 cm; for elderly/comorbid, recurrence after PN, hereditary multifocal RCC, informed preference. Tumours near hilum, bowel, pancreas or spleen → hydrodissection or laparoscopic mobilisation. No pathological staging after TA.
- Cryoablation: lethal ≤−20 °C, double freeze–thaw, iceball ~1 cm beyond tumour, real-time USG monitoring; radiofrequency has no visible equivalent. Neither superior.
- Higher local recurrence than PN (cryoablation 3–10%, radiofrequency 5–20%), mostly salvaged by repeat TA; no difference in metastasis or cancer-specific survival. Central/nodular enhancement in the ablation zone = recurrence → biopsy/retreat. Salvage surgery hampered by fibrosis.

Familial RCC — the 3-cm rule (National Cancer Institute)

- VHL solid tumours <3 cm: no metastases at ~5 y; ~25% metastasise when larger → intervene at **3 cm** with PN/enucleation/TA to preserve nephrons and delay dialysis. Applies to VHL, hereditary papillary, BHD. **HLRCC is the exception** — FH-deficient tumours metastasise early even when small, so excise at any size with wide margins.
- Recurrence is near-universal (occult microscopic disease) → lifelong surveillance; repeat PN is difficult, so TA helps. Belzutifan cuts interventions (renal response 30–50%; anaemia, fatigue).
- VHL histology: solid clear cell tumours plus cysts lined by carcinoma (incipient tumours) — hence recurrence and preference for enucleation, even for central tumours.
- Before renal surgery in VHL, **exclude phaeochromocytoma** and CNS haemangioblastoma. Transplant after removal of all renal tissue is safe.

8 · LOCALLY ADVANCED RCC

IVC tumour thrombus (venous involvement 4–10%; cure 45–70%)

Suspect with leg oedema, right-sided/non-collapsing varicocele, dilated abdominal veins, proteinuria, pulmonary embolism, right atrial mass or non-functioning kidney.

Level	Extent	T	Vascular control
Renal vein	Vein/branches	T3a	Standard RN
I	At renal vein ostium	T3b	Satinsky clamp/stapler; keep ≥50% lumen; patch if needed
II	Below hepatic veins	T3b	Sequential clamp: infrarenal IVC → contralateral renal vein → suprarenal IVC; occlude lumbars; extract in bloodless field

Level	Extent	T	Vascular control
III	Intrahepatic, below diaphragm	T3b	Mobilise liver, milk thrombus below hepatic veins; or suprahepatic clamp + Pringle; venovenous bypass often
IV	Above diaphragm	T3c	Cardiopulmonary bypass ± hypothermic circulatory arrest; avoid opening the heart if thrombus can be brought below the atrium
Any, wall invasion	Caval wall	T3c	Resect to negative margins, graft/reconstruct

- **Pre-op:** MRI or CT as close to surgery as possible (thrombus grows); enhancement distinguishes tumour from bland thrombus. **No routine renal artery embolisation** — no benefit, more complications and deaths. Coronary angiography before planned bypass (levels III–IV). Intra-op transoesophageal echo defines the tip, detects emboli, confirms clearance.
- **Sequence:** early arterial ligation, mobilise kidney until attached only by the vein, then thrombectomy (or handle thrombus first to reduce embolism).
- Completely occluded IVC may not need reconstruction (collaterals). Distal bland thrombus → remove, or ligate IVC above it/filter to prevent embolism.
- Level and wall invasion independently predict survival; level IV without metastasis is still curable.
- Complications: hypothermic arrest → coagulopathy, stroke, myocardial infarction; supradiaphragmatic mortality 5–10%. Surgery also palliates (prevents emboli; relieves oedema, ascites, cardiac dysfunction, haematuria) but select metastatic patients carefully.

Locally invasive (T4)

- T4 (<2% of surgical series) usually presents with pain from posterior abdominal wall, nerve root or paraspinal invasion. Differential: adrenocortical carcinoma, urothelial carcinoma, sarcoma, lymphoma.
- Only ~40% of suspected invasion is confirmed — large tumours metastasise more readily than they invade (e.g. upper-pole "liver invasion" with intact planes).
- En bloc resection (bowel, spleen, pancreas, abdominal wall) is the only cure; prepare with bowel preparation, splenectomy vaccination, steroid cover if both adrenals at risk. Prognosis still poor. Debulking rarely indicated (1-y survival 10–20%); neoadjuvant therapy as a "litmus test" for borderline resectability.

Lymph node dissection

- Only 30–43% of enlarged nodes are malignant; probability rises with size (~20% at 7 mm, ~90% at 30 mm). Node risk factors: tumour ≥10 cm, cT3/T4, grade 3–4, sarcomatoid, necrosis; ≥2 factors → >40% node-positive in a prospective series. 45% of positive nodes lie away from the hilum, so hilar sampling under-stages.
- No proven survival benefit (EORTC 30881) → LND for clinically suspicious nodes, mainly for staging; bulky nodes behave like metastatic disease but resect if feasible. Rare isolated nodal disease may be durably controlled.

9 · RECURRENCE, ADJUVANT THERAPY & FOLLOW-UP

Local recurrence	After RN	After PN	After TA
Rate	2–4%	1.4–10%	Cryo 3–10%, radiofrequency ablation (RFA) 5–20%
Site	Fossa, adrenal, vein stump, nodes	Remnant, usually away from bed	Ablation zone
Management	Only 20–40% isolated → full metastatic work-up; en bloc resection → 30–50% long-term cancer-free	Repeat PN, completion nephrectomy, TA, AS	Repeat TA, AS, salvage surgery

- After RN: small unifocal recurrence is favourable; multifocal implies distant spread. Recurrence reflects biology, not approach. Radiotherapy only palliates.
- Positive PN margin: linked to recurrence only in high-grade/high-stage disease → usually surveillance; completion nephrectomy only for gross involvement with a healthy contralateral kidney.
- Rationale for adjuvant therapy: 20–35% develop distant metastasis after "curative" surgery (occult micrometastases). Failed: interleukin-2, interferon, medroxyprogesterone, radiotherapy, vaccines, girentuximab, most TKIs. **Adjuvant pembrolizumab** is FDA-approved for high-risk RCC.
- Neoadjuvant TKI: 10–30% shrinkage; may enable PN or thrombectomy; ~20% serious events (wound healing). Metastatic therapy → *Ch 144*.

Follow-up (AUA risk groups)

Risk	Definition	Chest imaging
Low	pT1 grade 1–2	X-ray
Intermediate	pT1 grade 3–4 or pT2	X-ray
High	pT3	CT
Very high	pT4, pN1, sarcomatoid/rhabdoid, gross positive margin	CT

- Imaging from 3–6 mo, intensity rising with risk group; very high risk imaged most often and up to 10 y. Microscopic positive margin → move up one level.
- Every visit: history, examination, creatinine, eGFR, urinalysis for proteinuria → nephrology if progressive. Contrast CT/MRI preferred; USG may alternate after 2 y in low/intermediate risk. **30% of recurrences occur after 5 y** → shared decision beyond 5 y.
- Bone scan only for bone pain, ↑ alkaline phosphatase or suspicious imaging; acute neurological symptoms → urgent brain/spine CT/MRI; PET not routine.
- Metastasis found → define extent, refer to oncology; resect/ablate isolated or oligometastatic disease in selected patients. New local or retroperitoneal recurrence → metastatic work-up; urologist decides on surgery/ablation.
- Benign pathology → no routine imaging. After TA: contrast imaging within 6 mo, then intermediate-risk schedule.

10 · OTHER RENAL MALIGNANCIES

Tumour	Clues	Management
Sarcoma (1–2%)	**Leiomyosarcoma commonest** (50–60%); capsule/sinus origin, huge, no nodes, hypovascular	Wide en bloc excision, negative margins — first resection is the best chance; margin + grade = prognosis
Lymphoma	Non-Hodgkin; multiple masses (45%) commonest; massive/atypical adenopathy, splenomegaly; prior lymphoma → recurrence until proven otherwise	**Biopsy, avoid nephrectomy**; R-CHOP ± radiotherapy
Metastases	Commonest renal malignancy (12% at autopsy); lung commonest; multifocal, 5–30 HU	Biopsy; systemic/palliative; nephrectomy only for refractory bleeding
Carcinoid	**Horseshoe kidney (82×)**	Excision; good prognosis
Small cell	Usually advanced	Surgery + platinum
Primitive neuroectodermal tumour (PNET)	Ewing family; CD99; t(11;22)	Multimodal; poor
Adult Wilms	3% of Wilms; triphasic	Paediatric protocols; worse than children

Sarcoma

- Any feature of an RCC subtype excludes primary sarcoma (vs sarcomatoid RCC). Favour sarcoma: origin from capsule/sinus, huge mass without nodes, fat or bone, hypovascularity (except haemangiopericytoma — highly vascular, embolise pre-op).
- Why recur? Mesenchymal tumours lack natural barriers; the pseudocapsule is infiltrated, so tumour extends unseen. High grade → lung metastases, death within months; low grade → repeated local recurrence.
- Renal leiomyosarcoma > liposarcoma — reverse of retroperitoneum. Liposarcoma mimics AML; osteosarcoma (calcified) mimics staghorn.
- MRI planning, intraoperative margin assessment; chemotherapy (doxorubicin/ifosfamide, gemcitabine/docetaxel) disappointing; preoperative radiotherapy may reduce positive margins.

Lymphoma, leukaemia and metastases

- Lymphoma: clinically silent; primary renal rare (no native lymphoid tissue); ~90% haematogenous. Commonest pattern: multiple interstitial nodules that coalesce.
- Risk ↑ with immunosuppression, acquired immunodeficiency syndrome, autoimmune disease, graft-versus-host disease. B symptoms common; renal failure more often from hypercalcaemia or urate nephropathy (tumour lysis) than replacement.
- Management: percutaneous biopsy; at exploration send frozen section first; R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, prednisolone). Nephrectomy only for uncontrollable haemorrhage or rare primary renal lymphoma. Leukaemia: children, diffusely infiltrative.
- Metastases: almost all haematogenous; lung, breast, gastrointestinal, melanoma; isodense, enhance only 5–30 HU; biopsy confirms.

> **KEY POINT** Almost all non-RCC malignancies are **hypovascular**; the key decision is surgical (sarcoma, carcinoid) vs medical (lymphoma, metastasis — biopsy, don't resect). RCC nodes land interaortocaval (right) or para-aortic (left); nodes centred elsewhere suggest lymphoma. A solitary enhancing mass years after organ-confined non-renal cancer is more likely primary RCC.

11 · LANDMARK TRIALS

Trial	Design	Result	Take-home
EORTC 30904	RCT RN vs elective PN, ≤5 cm, normal contralateral	PN better function; 10-y overall survival better with RN (81 vs 76%)	Flawed; PN's functional gain may be smaller than believed
EORTC 30881	RCT RN ± LND	No survival benefit	Underpowered, low-risk (4% node+); LND for staging
S-TRAC	Sunitinib vs placebo, high-risk clear cell	Disease-free survival 6.8 vs 5.6 y; no overall survival gain	Only positive adjuvant TKI trial
KEYNOTE-564	Pembrolizumab vs placebo, high-risk	Disease-free survival HR 0.63	FDA 2021; only positive adjuvant ICI trial
DISSRM	Registry AS vs intervention, small masses	1 death/metastasis on AS; 24% delayed intervention	AS safe for small masses

12 · MCQ ONE-LINERS

1. Most accepted environmental risk factor for RCC: tobacco.
2. VHL gene 3p25; with elongin B/C targets HIF-α for degradation.
3. Main HIF isoform in clear cell RCC: HIF-2α; inhibitor belzutifan.
4. VHL type 2C: phaeochromocytoma only; type 1: no phaeochromocytoma.
5. Only oncogene familial RCC: MET (hereditary papillary, 7q31).
6. BHD: FLCN 17p11.2 — chromophobe/hybrid oncocytic tumours, pneumothorax.
7. HLRCC: FH 1q42; leiomyomas; excise at any size.
8. Papillary RCC: trisomy 7 and 17; clear cell: 3p loss, 5q gain.

9. Cup-like CA-IX, CK7+: clear cell papillary tumour (benign).
10. Medullary carcinoma: sickle cell trait, SMARCB1 loss.
11. Commonest paediatric RCC: Xp11 TFE3 translocation.
12. ISUP grade 4: sarcomatoid or rhabdoid differentiation.
13. Stauffer syndrome: interleukin-6; persistence after nephrectomy = residual tumour.
14. CT fat <−20 HU = AML; enhancement >15–20 HU = RCC-suspicious.
15. TNM 8th: pelvicalyceal invasion = T3a.
16. Caval wall invasion = T3c at any level.
17. VHL abdominal MRI from age 15, every 2 years.
18. Hypothermia for PN if ischaemia ≥25–30 min.
19. 30% of recurrences occur after 5 years.
20. Commonest renal sarcoma: leiomyosarcoma.
21. Commonest malignant renal neoplasm overall: metastasis (lung primary).
22. Renal carcinoid: horseshoe kidney, 82-fold risk.
23. Cryoablation lethal temperature: −20 °C or colder; iceball ~1 cm beyond tumour.
24. Renal leiomyosarcoma > liposarcoma; retroperitoneum: liposarcoma commonest.
25. Genetic counselling for all renal malignancy at age ≤46.

138 Urothelial Tumours of the Upper Urinary Tract and Ureter

Reading time ≈ 28 min (slow pace)

1 · EPIDEMIOLOGY

- Upper tract urothelial carcinoma (UTUC) = **5–10%** of urothelial cancers; incidence rising (ureteric tumours).
- Peak age 70–90; male:female ≈ **2:1**. In the Balkans UTUC is **40%** of renal neoplasms.
- Bilateral ≤ 5%; multifocal **10–20%**; concomitant carcinoma in situ (CIS) 11–36%; concurrent bladder cancer **17%**.
- At diagnosis ≈ **60% are invasive** and 7–9% metastatic.
- After UTUC treatment, the bladder recurs in **22–47%** and the contralateral upper tract in 2–6%.

> **KEY POINT** **Why UTUC presents invasive:** the upper tract urothelium looks like bladder urothelium but sits on a much thinner (mesodermal) muscle layer, so tumours reach muscle, renal parenchyma or periureteric fat earlier than in the bladder.

2 · RISK FACTORS

- **Smoking** – the most important modifiable factor; relative risk (RR) **2.5–7**, dose-dependent, ureter > pelvis. Stopping for > 10 yr cuts the risk by 60–70%.
- **Occupational "amino tumours"** – benzidine and β-naphthylamine (dyes, rubber, textiles, petrochemicals, coal); mean exposure ≈ 7 yr, latency up to 20 yr.
- **Chronic infection, stones, obstruction** – RR 1.5–2; linked to squamous cell carcinoma (SCC), less often adenocarcinoma.
- **Iatrogenic** – cyclophosphamide (via its metabolite acrolein), anthranoid laxatives used > 1 yr, pelvic radiotherapy.

Feature	Aristolochic acid (Balkan endemic nephropathy [BEN], Chinese herb nephropathy)	Arsenic (blackfoot disease)	Analgesic abuse (phenacetin)	Lynch syndrome (LS)
Setting	Rural Balkans; Belgium 1992–93 (slimming herbs); Taiwan/China	Well water in SW Taiwan and Chile	Chronic analgesic users	Families with a mismatch repair (MMR) mutation
Mechanism	Aristolactam–DNA adducts → **A:T→T:A transversion** (TP53 codon 139)	Arsenic; causal link unproven for UTUC	Papillary necrosis; **capillarosclerosis** (pathognomonic)	MMR deficiency (MSH2 carries the highest risk)
Sex / age	No sex bias; ≈ 10 yr earlier than sporadic	Female 2:1; age 55–60	Latency ≈ 22 yr	< 60 (mean 55); relatively more women
Site	Bilateral more often	Ureter 2:1 over pelvis	Pelvis > ureter; look for contralateral disease	Upper tract > bladder
Inherited?	**Familial, not inherited** (shared diet)	No	No	Yes, autosomal dominant

> **KEY POINT** All four behave as a **field disease**, with more multiple and bilateral recurrences. That matters for treatment: kidney-sparing options become more attractive, but BEN patients often also have renal insufficiency. An analgesic abuse history is present in 22% of pelvic and 11% of ureteric tumours.

Lynch syndrome (hereditary nonpolyposis colorectal cancer, HNPCC)

- Caused by mutations in the MMR genes MLH1, MSH2, MSH6 and PMS2. UTUC risk is ≈ **14×** that of the general population and ≈ **75×** in MSH2 carriers. A germline mutation is found in **5–9%** of screened UTUC patients.
- **Screen at the first interview** (European Association of Urology, EAU) if any of: age < 60; a personal history of an LS-spectrum cancer; a first-degree relative (FDR) < 50 with an LS cancer; two FDRs with an LS cancer.
- Suspected cases → germline DNA sequencing → surveillance for other LS cancers (colorectal, endometrial) → family counselling. A sporadic-looking tumour with loss of MMR proteins on immunohistochemistry (IHC) also goes to germline testing.

3 · PATHOLOGY & MOLECULAR BIOLOGY

- > 90% of tumours are urothelial carcinoma (UC). Variant histology occurs in ≈ **25%** (squamous, glandular, sarcomatoid, micropapillary, neuroendocrine, lymphoepithelial) and marks high-grade, aggressive disease. **Micropapillary** is an independent predictor of worse outcome.
- Typical sequence: hyperplasia → dysplasia → CIS. CIS is easily missed (whitish plaque or red patch).
- Ureteric inverted papillomas harbour malignancy in ≈ **18%**, so follow them for ≥ 2 yr.
- SCC is 0.7–7% of tumours, 6× commoner in the pelvis, and invasive at presentation. Adenocarcinoma is < 1%, linked to stones and obstruction, with a poor prognosis.
- Ureteric tumours are **70% distal**, 25% mid and 5% proximal, reflecting downstream implantation. Sessile tumours invade in > 80%.
- **Molecular** – UTUC and bladder cancer are genetically distinct but usually clonally related in the same patient. Of five molecular subtypes, **RAS-mutated** has the worst prognosis. Microsatellite instability (MSI) and hypermethylation are the key differences from bladder cancer; MSI is independently prognostic and flags LS. No tissue marker is in routine use.

4 · CLINICAL FEATURES & DIAGNOSIS

- **Haematuria** is the commonest sign (56–98%). Flank pain comes next (20–32%): dull from gradual obstruction, or colicky as clots pass. A mass is found in ≈ 10% and ≈ 15% are incidental. Systemic symptoms → metastatic work-up.
- **Cystoscopy is always done** because bladder cancer often coexists. Positive cytology with normal cystoscopy (including the urethra) → evaluate the upper tracts.

Test	Role and key points
Computed tomography urography (CTU) – first line; the EAU prefers it to MR urography	Sensitivity ≈ **92%**, specificity ≈ 95%. Signs: filling defect (50–75%), obstruction, non-visualisation. Misses flat lesions and invasion of small tumours. UC ≈ **46 Hounsfield units (HU)** vs ≈ 100 HU for a uric acid stone.
Hydronephrosis	Means an invasive ureteric tumour in up to 80%; an independent adverse predictor.
Magnetic resonance (MR) urography	Used when CT contrast or radiation is contraindicated. Sensitivity ≈ 75% for tumours < 2 cm. Avoid gadolinium if creatinine clearance < 30 mL/min (risk of nephrogenic systemic fibrosis).
Retrograde pyelography	Accuracy ≈ 75%.
Fluorodeoxyglucose positron emission tomography/CT (FDG-PET/CT)	Nodes/metastases when iodinated contrast is contraindicated.
Split-function renal scan	Always assess contralateral function before radical surgery.
Flexible ureteroscopy (URS) + biopsy	Raises diagnostic accuracy from 75–85% to ≈ 90%. Biopsy gives the grade in ≈ 90%; visual grading is right in only ≈ 70%.

Test	Role and key points
Urine cytology	Sensitivity ≈ 65%, depending on grade: G1 20%, G2 45%, G3 75%. Selective ureteric washing beats voided urine; brush biopsy + cytology reaches ≈ 90%. **Collect before contrast**, which degrades the cells.
Fluorescence in situ hybridisation (FISH)	More sensitive than cytology; best for low-grade surveillance after kidney-sparing treatment; not routine.

> KEY POINT **Why URS biopsy cannot stage:** samples are small and rarely include deep layers, so a Ta biopsy is often upstaged to T3 at resection. Stage is therefore **inferred** from grade, appearance, hydronephrosis and cytology, and patients treated with kidney-sparing surgery need intensive follow-up.

- **Indications for diagnostic URS:** diagnostic uncertainty; kidney-sparing treatment under consideration; solitary kidney. **Pitfall:** URS before radical nephroureterectomy (RNU) increases intravesical recurrence, especially with pelvic tumours and biopsy, although survival is unchanged. Avoid it when CTU and cytology already justify RNU.

5 · STAGING & GRADING

Tumour, node, metastasis (TNM, 8th ed.)	Definition
Ta / Tis	Non-invasive papillary / CIS
T1	Subepithelial connective tissue
T2	Muscularis
T3	Pelvis: peripelvic fat or **renal parenchyma**; ureter: periureteric fat. pT3a = microscopic parenchymal invasion; pT3b = macroscopic invasion or peripelvic fat (worse)
T4	Adjacent organs, or through the kidney into perinephric fat
N1 / N2	Single node ≤ 2 cm / single node > 2 cm or multiple nodes

- American Joint Committee on Cancer (AJCC) stage: 0a/0is; I = T1; II = T2; III = T3; **IV = T4, any N+ or M1**.
- Nodes: pelvis/upper ureter → hilar and retroperitoneal; lower ureter → iliac/pelvic.
- Grade by World Health Organization (WHO) 2004/2016: papillary urothelial neoplasm of low malignant potential (PUNLMP), low grade, high grade. PUNLMP is managed as low grade.

6 · PROGNOSTIC FACTORS & RISK STRATIFICATION

- **Stage is the single most important factor**, then grade. 5-yr cancer-specific survival (CSS) in high-risk RNU cases (Surveillance, Epidemiology and End Results [SEER] data): ≤ T1 86%, pT2 77%, pT3 63%, pT4/N+ 39%.
- **Ureteric and multifocal tumours do worse than pelvic tumours** at the same stage. The renal parenchyma acts as a barrier, whereas periureteric tissue opens straight into a rich vascular and lymphatic plexus.
- **Preoperative factors:** age (predictive, but never a reason by itself to deny curative RNU); smoking; high American Society of Anesthesiologists (ASA) score and poor Eastern Cooperative Oncology Group performance status (ECOG-PS); obesity; adverse blood markers (low albumin, high neutrophil-to-lymphocyte ratio, high C-reactive protein); **surgical delay – do RNU within 12 wk** for invasive disease.
- Size: **2 cm** is the best cut-off for predicting ≥ pT2.

- **Postoperative factors:** nodal metastasis (up to 40%) and extranodal extension; lymph node density ≥ 30%; lymphovascular invasion (LVI, ≈ 20%; independent in node-negative disease and never seen on biopsy, so it must be reported on every specimen); positive margins; necrosis > 10%; sessile architecture; concomitant CIS; variant histology.

Low risk — ALL required	High risk — ANY ONE (EAU 2020)
Unifocal	Multifocal
< 2 cm	> 2 cm
Low-grade cytology	High-grade cytology
Low-grade URS biopsy	High-grade URS biopsy
No invasive aspect on CTU	Hydronephrosis
—	Previous radical cystectomy
—	Variant histology

KEY POINT Because true stage is known only after RNU, every preoperative factor is a stand-in for invasion. The rule is asymmetric: a single adverse feature makes the tumour high risk. **Trap:** a small, low-grade unifocal tumour **with hydronephrosis is high risk**.

7 · RECURRENCE PATTERNS

- **Bladder after UTUC:** 15–75% at 5 yr, usually within 2 yr.
- **Upper tract after bladder cancer:** 2–4%, late (median > 3 yr); 0.75–6.4% after cystectomy. About **30%** of patients with bladder CIS treated with bacillus Calmette–Guérin (BCG), or cystectomised for BCG-refractory disease, develop UTUC. Risk factors: distal ureter/orifice tumour, multifocal or recurrent non-muscle-invasive disease, CIS, reflux.

KEY POINT **Why recurrence is asymmetric:** urine flows downhill. Tumour cells seed antegrade, so bladder recurrences are common, early and monoclonal. The field effect (independent clones in carcinogen-exposed urothelium) explains multifocal tumours of different clonal origin; both operate.

Patient factors	Tumour factors	Treatment factors
Male, prior bladder cancer, smoking, chronic kidney disease (CKD)	Positive cytology, ureteric location, multifocal, invasive pT stage, necrosis	Laparoscopic approach, extravesical cuff removal, positive margins, diagnostic URS

Predictors of intravesical recurrence after RNU

- **Pan-urothelial disease** = bladder plus ≥ 2 extravesical sites (women: both upper tracts; men: upper tract(s) and/or the prostatic urethra). There is no evidence-based algorithm. Cystectomy is used for high-grade bladder disease, conservative treatment for non-infiltrating UTUC, and total urinary tract removal is an option in younger patients. Cystectomy followed by RNU loses renal function and can make the patient ineligible for cisplatin.

8 · MANAGEMENT OVERVIEW & FOLLOW-UP

ALGORITHM

Cystoscopy + CTU ± URS biopsy/cytology + contralateral function → LS screening if indicated → risk-stratify

Low risk → kidney-sparing treatment (endoscopic or segmental) with intensive surveillance

High risk → RNU with bladder cuff ± lymph node dissection (LND), within 12 wk

Clinically node-positive (cN+) → systemic chemotherapy first → surgery if there is a response

After RNU → single postoperative intravesical chemotherapy dose → cystoscopic follow-up for everyone

- Preoperative nomograms predict ≥ pT2 or non-organ-confined disease and are used to select patients for neoadjuvant chemotherapy; pathological downstaging improves overall survival. Perioperative chemotherapy (neoadjuvant, POUT) → *Ch 139*.
- **Why give chemotherapy before surgery?** RNU removes a kidney, and the resulting loss of renal function often makes patients ineligible for cisplatin afterwards.
- LND at RNU gives the best staging; whether it cures is debated.
- A single postoperative intravesical instillation reduces bladder recurrence after RNU.
- Surgical technique (RNU steps, distal ureter management, endoscopic ablation, segmental ureterectomy, topical instillation) → *Ch 139*.
- **Follow-up:** cystoscopy for all patients. After kidney-sparing treatment, follow intensively with URS, in-situ cytology and CTU. After high-risk bladder cancer (BCG-treated CIS, BCG-refractory disease), check the upper tracts every year with CT, which should replace cytology and excretory urography. The risk stays constant, so surveillance is prolonged.

9 · MCQ ONE-LINERS

1. UTUC = 5–10% of urothelial cancers; Balkans: 40% of renal neoplasms.
2. Smoking RR 2.5–7; ureter at greater risk than pelvis.
3. Lynch: UTUC risk 14×; MSH2 carriers 75×.
4. RAS-mutated molecular subtype has the worst prognosis.
5. Aristolochic acid signature: A:T→T:A transversion, TP53 codon 139.
6. BEN: familial, not inherited; bladder cancer not increased in affected families.
7. Arsenic UTUC: blackfoot disease; women 2:1.
8. Phenacetin latency ≈ 22 yr; capillarosclerosis is pathognomonic.
9. Ureteric inverted papilloma: malignancy ≈ 18%; follow ≥ 2 yr.
10. CTU: sensitivity 92%, specificity 95%.
11. UC ≈ 46 HU; uric acid stone ≈ 100 HU.
12. Cytology sensitivity: G1 20%, G2 45%, G3 75%.
13. Pelvic tumour invading renal parenchyma = T3, not T4.
14. N1 = single node ≤ 2 cm; any N+ = AJCC stage IV.
15. Necrosis > 10% is an independent adverse predictor.
16. Lymph node density ≥ 30% predicts poor outcome.
17. Uroplakin III and p27 expression are favourable.

139 Surgical Management of Upper Urinary Tract Urothelial Tumours

Reading time ≈ 46 min (slow pace)

Epidemiology, risk factors, pathology, staging and imaging → *Ch 138*

1 · PRINCIPLES & PATIENT SELECTION

- **Standard therapy** for upper tract urothelial carcinoma (UTUC): radical nephroureterectomy (RNU) with bladder cuff, provided the contralateral kidney is normal and the patient is fit.
- UTUC is **unforgiving of spillage**: it implants wherever tumour-laden urine goes, so every approach keeps the collecting system closed.
- **Work-up**: computed tomography (CT) or magnetic resonance imaging (MRI) urogram → cystoscopy → ureteroscopy (URS) with biopsy and selective cytology. URS raises diagnostic accuracy from ≈75% (urography) to 85–90%, shows lesions missed on imaging and is essential when diagnosis is uncertain or kidney-sparing management (KSM) is planned. Retrograde access impossible (e.g., urinary diversion) → percutaneous antegrade endoscopy; endoscopy unsafe → CT- or ultrasound-guided needle biopsy.
- Diagnostic endoscopy itself carries risk (bladder seeding, pyelovenous/pyelolymphatic migration), so keep it brief and low-pressure; diagnosis and treatment may be combined for small accessible tumours.
- URS biopsies are tiny and shallow, so they cannot stage. Stage is **inferred** from grade, endoscopic appearance, imaging and cytology; low grade (LG) tumours are mostly Ta/T1, high grade (HG) mostly invasive.

American Urological Association / Society of Urologic Oncology (AUA/SUO) risk group	Features	Ablation	Systemic therapy
Low – favourable	LG, negative cytology, unifocal, papillary, no obstruction/invasion	**Preferred**	No
Low – unfavourable	LG + obstruction, multifocality or prior lower-tract tumour	May be offered	No
High – favourable	HG, unifocal, papillary, no invasion/obstruction	Rare, select cases	Neoadjuvant/adjuvant
High – unfavourable	HG cytology, radiographic invasion, obstruction, suspicious nodes, multifocal, sessile/flat	Palliation only	Neoadjuvant/adjuvant

> **KEY POINT** Low grade does not automatically mean ablation: obstruction or multifocality moves an LG tumour into the unfavourable group, where RNU retains a role.

> **KEY POINT** The algorithm balances two axes — tumour risk and renal reserve. High risk with adequate reserve goes to extirpation, low risk to preservation; imperative indications are where the two conflict.

ALGORITHM

Imaging suspicion → CT/MR urogram → endoscopy + biopsy + cytology → risk group + shared decision

High risk + adequate glomerular filtration rate (GFR) → RNU ± neoadjuvant chemotherapy · Low risk or poor GFR → kidney-sparing

- **Elective KSM**: low-risk disease, irrespective of the contralateral kidney.

- **Imperative KSM** (selected high risk): chronic kidney disease (CKD), solitary kidney, bilateral tumours, unfit patient. Treatment must not significantly compromise cancer control; dialysis risk is weighed against oncological risk by shared decision.
- **Lower ureteric** tumours, especially unfavourable → distal ureterectomy rather than endoscopy.
- **Expectant management** (AUA/SUO option): only with major comorbidity, competing mortality or high end-stage renal disease (ESRD) risk; untreated UTUC causes progression, pain, haematuria and obstruction, with markedly worse survival.

2 · ENDOSCOPIC (KIDNEY-SPARING) MANAGEMENT

- Principle as in the bladder: sample, then ablate with cautery or laser. **Grade is the key selector** and the strongest predictor of recurrence; results for HG are poor.
- AUA/SUO: endoscopic treatment for tumours **<1.5 cm**; caution ≥1.5 cm (invasion risk).
- **Staged approach** for bulky or suspected HG tumour: biopsy and partial ablation only, so that perforation is avoided if RNU follows; complete treatment once pathology confirms suitability. Unresectable, HG or invasive → RNU.
- Versus RNU: similar survival but **~3× more reinterventions** (upper-tract recurrence). Recurrence is expected, several sessions may be needed, and the patient must accept lifelong endoscopic and radiographic surveillance.
- Difficult sites: tumours of the renal papillae or fornix are hard to treat without perforation, bleeding or functional loss. HG or invasive tumours suit endoscopy only in imperative settings.

	Retrograde ureteroscopic	Antegrade percutaneous
System	**Closed** – non-urothelial surfaces not exposed	**Open** – tract seeding ≈0.3%, all HG
Best for	Low-volume ureteric and small renal tumours	Large pelvis/upper ureteric tumours, lower-pole calyx, urinary diversion
Staging	Not possible	Possible – base biopsied separately
Morbidity	Low	Higher (≈25%), admission
Recurrence	Higher bladder and upper-tract	Lower
Second look / topical Rx	Repeat URS; catheter or stent reflux	Via retained nephrostomy

Ureteroscopy – steps

- Cystoscopy → cytology → wire or no-touch access → 6.9–7.5 Fr semirigid scope → flexible scope for all calyces → map lesions → saline barbotage improves cytology yield.
- Inspect the orifice for lateralising haematuria and the distal ureter for wire trauma before advancing under vision.
- Tight ureter: gentle minimal dilation or pre-stent for interval URS (passive dilation) rather than forceful dilation.
- Access sheath — no recommendation. For: low pressure, repeated access, larger specimens, lower ureter bypassed (less seeding). Against: extra dilation, less scope flexibility, post-op stent, ureteric injury.
- Specimens are minute: fix at once and label for histology or cytology. Adding selective cytology to biopsy improves grading accuracy.
- Techniques: bulk excision with forceps or flat wire basket + base ablation (LG on narrow stalk; avoid avulsing the ureter); ureteroscopic resection of intraluminal tumour only, never beyond lamina propria, because the mid/upper ureter is thin; **biopsy (3-Fr forceps) then laser** — the usual approach, as laser depth is more reliable.
- Tumour protruding from the orifice (intramural ureter) → complete ureteroscopic ablation or transurethral resection of the whole distal ureter.

- Bugbee electrode: unpredictable depth; **never fulgurate circumferentially** (stricture). Post-ablation stent at surgeon discretion; pre-stenting in difficult access reduces injury and spillage.

Laser (yttrium-aluminium-garnet, YAG)	Penetration	Role
Holmium (Ho:YAG)	<0.5 mm	Ideal for ureter
Neodymium (Nd:YAG)	5–6 mm	Transmural injury risk in ureter
Thulium	<0.2 mm	Near water-absorption peak; strong coagulation

Percutaneous – steps

- Cystoscopy → open-ended ureteric catheter → contrast to define calyces → puncture (prone split-leg allows flexible URS guidance). Peripheral calyceal tumour: puncture the calyx distal to it, never through it. Pelvis/upper ureter: upper or middle calyx.
- Dilate (Amplatz or balloon) to 30 Fr; option to let the tract mature 1–2 weeks first. Diagnostic-only cases need just a small tract for a flexible scope.
- Nephroscope in → ureteric catheter pulled out through the tract and exchanged for a stiff wire (antegrade + retrograde control) → rigid and flexible nephroscopy; suspected upper ureteric tumour → antegrade URS.
- Cold-cup forceps (LG, thin stalk; base cauterised), resectoscope for broad-based tumours (**bipolar** — monopolar risks hypo-osmotic irrigant absorption; avoid major vessels) or laser. Base sent separately for staging.
- **Nephrostomy always left** → second-look nephroscopy and route for topical therapy. HG or invasive histology → RNU.
- Urinary diversion: ureteroenteric anastomosis is hard to find; know the anastomosis type (Wallace vs Bricker conduit, neobladder). Try flexible endoscopy first; if ureters not found → antegrade wire into conduit/pouch, retrieve, then retrograde URS over it; failing that, full tract dilation and antegrade treatment.

> **KEY POINT** Does URS spread tumour? Prior ureteroscopic manipulation increases bladder recurrence (downstream seeding), but pre-RNU URS has not increased metastasis. Mitigate with a single postoperative intravesical instillation.

3 · SEGMENTAL URETERECTOMY

- Preserves nephrons **and** yields a specimen that guides adjuvant therapy. In selected patients outcome equals RNU: stage and grade, not extent of surgery, set prognosis.
- **Ureteroscopy of the entire tract above the tumour is mandatory** to exclude multifocality. HG → add ipsilateral pelvic lymph node dissection (LND). Resection for poor renal function must not sacrifice oncological efficacy merely to avoid dialysis.
- Ipsilateral recurrence after conservative ureteric surgery **33–55%**, mostly distal to the original site, and lifelong.

Procedure	Indication	Key steps
Segmental resection + uretero-ureterostomy	LG proximal/mid tumour not ablatable (size, number); HG when unit must be saved. Rare.	Ligate 1–2 cm above and below; frozen-section margins; spatulate; 4-0 polyglactin; stent; tension-free with preserved blood supply
Distal ureterectomy + reimplant	LG low-stage, or selected HG/invasive distal tumour when renal preservation needed	Whole distal ureter + cuff; 2-layer posterior cystotomy closure; ± LND

- **Uretero-ureterostomy technique**: flank incision from tip of 12th rib → extraperitoneal → ureter slung in vessel loops → tumour palpated. For a long gap, mobilise and descend the kidney.
- Reimplant: **non-refluxing** confines infection and theoretically prevents upper-tract seeding; **refluxing** eases later upper-tract surveillance. Debated.

- **Extravesical reimplant**: incise detrusor → mucosal slit distally → full-thickness ureter-to-mucosa anastomosis (3-0 absorbable) → two anchoring sutures through bladder wall → close detrusor over ureter (non-refluxing). **Intravesical**: anterior cystotomy → 2–3-cm submucosal tunnel.
- **Psoas hitch**: mobilise bladder; divide round ligament (female) and, if needed, contralateral superior vesical pedicle; suture ipsilateral dome to psoas tendon avoiding nerve entrapment.
- **Boari flap**: preoperative cystogram; small or irradiated bladder contraindicates; base ≥2 cm wider than apex (blood supply); width ≥3× ureteric diameter (tube calibre); U-shaped (L-shaped for more length); tip fixed to psoas; tubularised in two layers over a catheter.
- **Ileal ureter**: 20–25 cm ileum ≥15 cm from ileocaecal valve, isoperistaltic; proximal end to renal pelvis, distal end-to-side to posterior bladder; large Foley ≥1 week (irrigate mucus); cystogram and nephrostogram before tube removal.
- Laparoscopic/robotic distal ureterectomy + reimplant: reserved for low-risk distal tumours, same oncological rules.
- Post-op: drain, Foley 5–7 d, cystogram before removal after extensive reconstruction.

Manoeuvre	Length gained	Condition
Psoas hitch	≈5 cm	Mobile bladder; avoid nerve entrapment
Boari flap + hitch	Further 10–15 cm	Adequate capacity
Renal descensus + nephropexy	Up to 8–10 cm	Better on left (longer renal vein)
Ileal ureter / autotransplant	Whole ureter	Autotransplant in skilled hands

4 · POSITIVE CYTOLOGY & CARCINOMA IN SITU (CIS)

ALGORITHM

Positive voided cytology → repeat → CT urogram + bladder biopsies → bladder positive: treat bladder

Persistent positive with negative bladder (or after bacillus Calmette–Guérin (BCG) "6+1") → selective upper-tract cytology + retrograde pyelography (RPG) + prostatic urethral biopsy (men) + URS

- Extravesical evaluation needs **selective** cytology from each upper tract without bladder or urethral contamination.
- Upper-tract CIS is a diagnosis of exclusion (persistent positive selective cytology, normal URS and imaging); narrow-band or blue-light imaging and brush biopsy help. **RNU for cytology alone is not recommended**: cytology gives false positives (infection, stones) and contralateral disease often follows. Segmental resection fails (field change).
- AUA/SUO: **6-week BCG induction**, preferably via nephrostomy. Initial response is good, but late recurrences can be lethal, so follow-up must be vigilant. RNU if more than surface disease is proven. Retrograde alternative: weekly cystoscopy with ureteric catheter in the upper calyx.
- CIS coexisting with HG papillary UTUC → radical, possibly multimodal treatment (worse survival).
- Re-evaluate frequently: urinalysis, bladder ± selective cytology, cystoscopy, RPG or URS.
- CIS at the ureteric margin during cystectomy: frozen section; converting to a negative margin is achieved in most and associated with better survival.

5 · TOPICAL THERAPY IN THE UPPER TRACT

- Indications: solitary kidney, bilateral UTUC, ESRD risk. Liquid agents drain quickly by gravity, so delivery route determines efficacy.
- Routes: antegrade via nephrostomy (gravity, ≤25 cm H_2O column); retrograde ureteric catheter; reflux via indwelling stent.
- **Always low pressure and sterile urine**: high pressure and infection cause sepsis and systemic absorption (deaths reported).

- Agents as in the bladder: thiotepa, MMC, gemcitabine, docetaxel, epirubicin, doxorubicin, BCG — evidence only small uncontrolled series.
- Exclude active infection before **every** instillation.
- Adjuvant instillation after papillary ablation has **no proven benefit**; adjuvant BCG acceptable after complete ablation of HG favourable Ta/T1.
- Complications: sepsis, stricture (from prior endoscopy, catheterisation or the agent); BCG – renal granulomas (commonest), disproportionate function loss with reflux.
- **UGN-101 (Jelmyto)**: mitomycin C (MMC) in reverse-thermal hydrogel — liquid when instilled, gels at body temperature and dissolves over hours, prolonging contact. US Food and Drug Administration (FDA) approved 2020 for LG UTUC **not endoscopically manageable**. Toxicity: ureteric stricture ≈44% (fewer antegrade), marrow suppression, embryo-fetal toxicity. Complete endoscopic ablation remains the mainstay; offer gel after disclosing toxicity.

6 · RADICAL NEPHROURETERECTOMY

Indications

- Non-metastatic high-risk tumour of renal pelvis or proximal ureter (HG biopsy/cytology, obstruction, parenchymal invasion), with bladder cuff and LND.
- LG with unfavourable features (obstruction, multifocal, large volume) or failed/infeasible endoscopy.

Open RNU

- Supine or modified flank; genitalia in field for catheter access. Midline: best for nodes and bladder, poor upper pole (left, obese). Flank, subcostal or thoracoabdominal incisions need a second Gibson or Pfannenstiel incision for the cuff.
- **Kidney first, cuff last**: line of Toldt → colon medial → **clip ureter early** (stops tumour cells being milked into the bladder) → control hilum before manipulation → artery then vein → kidney outside Gerota's fascia.
- Hazards: duodenum (right); pancreatic tail, spleen (left). **Adrenalectomy not routine** — only if involvement suspected.
- If ureter divided (clips/ties/stapler), **never seal it with a vessel-sealer alone**.

Distal ureter & bladder cuff

- Remove the **entire intramural ureter and orifice**: a retained stump recurs in **30–75%**, cannot be practically surveyed, and worsens survival. After bowel diversion, excise the ureteroenteric anastomosis with a bowel cuff.
- Principles: excise orifice with small urothelial margin → contain spillage → watertight closure.

Technique	Status
Extravesical (laparoscopic/robotic; open with clamp)	Acceptable. Hazards: entering the tunnel, contralateral orifice
Transvesical (open cystotomy)	Acceptable; 1-cm mucosal cuff
Transvesical "lasso"	Abandoned – spillage, recurrence
Endoscopic "pluck"	Abandoned – extravesical spillage
Intussusception "stripping"	Abandoned – recurrence
Laparoscopic "staple"	Abandoned – urothelium trapped in staple line → positive margins

- **Extravesical steps**: follow ureter under superior vesical artery → tie at detrusor → partly fill bladder → divide lateral pedicle → empty bladder → excise cuff en bloc.
- Minimally invasive: open bladder supralaterally so it drains without spillage; flexible cystoscopy confirms complete excision.

- Closure: two layers absorbable (mucosa, then detrusor/adventitia); Foley several days; perivesical drain optional.

Laparoscopic & robotic RNU

- Specimen extracted **intact in a bag** via incision that also serves the distal ureterectomy.
- Steps: Foley before insufflation; camera at umbilicus; colon reflected medially, leaving lateral Gerota attachments so the kidney does not fall onto the hilum → early ureteric clip → artery then vein (stapler or clips) → wide perihilar margin (lymphatic spread).
- Extraction incision: Pfannenstiel/low midline if dissection reached below the iliacs; Gibson if more proximal ureter needs exposure.
- **Laparoscopic distal ureter**: Trendelenburg → free ureter below iliacs → divide vas or round ligament → trace under superior vesical artery → clip at detrusor → divide lateral pedicle → empty bladder → excise cuff under vision. Robotic: newer systems need no re-docking.
- Ureteric blood supply: anteromedial (upper ⅓), medial (middle ⅓), **lateral** (lower ⅓).
- Relative contraindications: bulky tumour invading neighbours; extended post-chemotherapy LND.

	Open	Laparoscopic	Robotic
Oncology	Reference	Equivalent	Comparable
Morbidity	Highest	Less blood loss, shorter stay	Lowest; fewer positive margins
Caveat	—	Possibly more bladder recurrence; less favoured for T3/T4	Longest; more LND done

> **KEY POINT** Approach matters less than principle: port-site seeding and peritoneal carcinomatosis follow spillage, not the approach.

Lymphadenectomy

- Staging, prognostic and therapeutic value in **T2–T4**, especially renal pelvis/proximal ureter.
- Removing **≥8 nodes** improves detection and disease-specific survival above pT1. Surgeon, not approach, determines yield. Outcome: pN0 > pNx > pN+. Specific complication: chylous leak.

Primary	Right	Left	Caudal limit
Renal pelvis	Hilar, paracaval, retrocaval, **interaortocaval**	Hilar, para-aortic	Inferior mesenteric artery
Upper ⅔ ureter	As renal pelvis		Aortic bifurcation
Lower ⅓ ureter	Ipsilateral iliac, obturator, presacral		Pelvis

Lymph drains right → left, so right-sided tumours need interaortocaval nodes; on the left they add little.

Results

- Prognostic cliff between **T2 and T3**; HG behaves like T3; node-positive disease with surgery alone behaves almost like metastatic disease — the case for systemic therapy.
- Bladder recurrence (drop-down implantation) is common; loss of renal function is RNU's main cost.

7 · PERIOPERATIVE ADJUNCTS

Single-dose intravesical chemotherapy

- Bladder tumours develop in 30–50% after RNU or endoscopic treatment, mostly within 2 years, from seeding. **AUA/SUO strongly recommend** one dose after RNU (optional after endoscopy).
- Postoperative (3–7 d): cystogram excludes cuff leak → MMC or gemcitabine → dwell 1–2 h.
- **Intraoperative** (authors' preference): two- or three-way catheter → drain → instil before surgery → clamp 1–2 h with minimal intravenous fluid → unclamp once ureter is clipped, so the bladder is empty before cuff dissection. Avoids cystogram and leak-related cancellation; drug present when seeding occurs.

Perioperative systemic chemotherapy (high risk)

- **Renal function is the hinge**: CKD is common and worsens after RNU, so many lose cisplatin eligibility — hence give platinum **before** the kidney is removed.
- **Neoadjuvant**: AUA/SUO standard option for high-risk UTUC, especially if post-op creatinine clearance (CrCl) expected **≤50** mL/min. EA8141: CrCl >50 → accelerated MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) ×4; 30–49 → gemcitabine–carboplatin; pathological complete response 14%, <ypT2N0 60%.
- **Adjuvant** (POUT): pT2–T4 or N+ → gemcitabine–cisplatin (CrCl >50) or –carboplatin (30–49). Less effective with carboplatin and N+ → cisplatin preferred.
- Regional nodes on imaging → chemotherapy first, then consolidative surgery.

8 · COMPLICATIONS & THEIR MANAGEMENT

Procedure	Complication	Management / prevention
URS	Perforation; stricture	Stent; pre-stent tight ureter; avoid circumferential fulguration
Percutaneous	Bleeding/transfusion, irrigant absorption, perforation, secondary ureteropelvic junction obstruction	Bipolar resection; avoid major vessels; nephrostomy
Topical therapy	Sepsis, absorption, stricture	Sterile urine, low pressure
Cuff closure	Urine leak	Two-layer closure; catheter; cystogram
RNU	Renal insufficiency	Nephrology referral

9 · FOLLOW-UP

- Every visit: history, examination, metabolic panel, **cystoscopy**, cytology; after KSM add ipsilateral URS. Do not routinely scope a normal contralateral tract.
- CT/MRI urography (bilateral metachronous risk); RPG if contrast impossible. Filling defects need URS. Chest imaging for metastases.
- Refer to nephrology if renal function falls after RNU; advise smoking cessation.

Group	Cystoscopy	Upper-tract endoscopy	Imaging
KSM low risk	3 mo, then 6–9-monthly ×2 yr, then yearly	3 mo, 6 mo, 1 yr	6–9-monthly ×2 yr, yearly to 5 yr
KSM high risk	3 mo, then 3–6-monthly ×3 yr, then yearly	3 mo, 6 mo, 1 yr	3–6-monthly ×3 yr, yearly to 5 yr
RNU <pT2N0	+ cytology at 3 mo, then by grade	—	CT/MRI by 6 mo, yearly ≥5 yr
RNU ≥pT2	3 mo, 3–6-monthly ×3 yr, yearly	—	Abdomen/pelvis 3–6-monthly ×2 yr; chest CT 6–12-monthly ×5 yr

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
ODMIT-C	Randomised controlled trial (RCT), MMC 40 mg single dose at catheter removal after RNU	Bladder recurrence 28% → 16% at 1 yr	Single-dose instillation is standard

Trial	Design	Result	Take-home
Pirarubicin trial (Ito)	RCT, single pirarubicin dose within 5 days of RNU; recurrence biopsy-proven	Bladder recurrence 42% → 17% at 2 yr	Second RCT confirming single post-RNU instillation; different drug, same benefit
Laparoscopic vs open RNU	Only RCT; small	Equal 5-yr survival; less blood loss	Minimally invasive acceptable
UGN-101 phase III	Single-arm, LG UTUC	Complete response 59%; 56% disease-free at 12 mo	Gel chemoablation option
POUT	Phase III adjuvant platinum vs surveillance, pT2–4/N+	Stopped early for efficacy; disease-free survival (DFS) hazard ratio **0.45**; 3-yr DFS 71% vs 46%	Adjuvant platinum standard after RNU for ≥pT2/N+

11 · MCQ ONE-LINERS

1. Retained intramural stump recurrence: 30–75%.
2. AUA/SUO endoscopic size cut-off: <1.5 cm.
3. Most important predictor of recurrence after endoscopic treatment: grade.
4. Distal-third ureteric blood supply is lateral.
5. Ho:YAG penetration <0.5 mm; Nd:YAG 5–6 mm; thulium <0.2 mm.
6. Percutaneous tract dilated to 30 Fr; tract seeding ≈0.3%, all high grade.
7. Psoas hitch ≈5 cm; Boari flap further 10–15 cm.
8. Boari: width ≥3× ureteric diameter; base ≥2 cm wider than apex.
9. Ileal ureter: 20–25 cm ileum, ≥15 cm from ileocaecal valve.
0. ≥8 nodes removed improves disease-specific survival above pT1.
1. Right renal pelvis template includes interaortocaval nodes.
2. Renal pelvis template caudal limit: inferior mesenteric artery.
3. Bladder recurrence after RNU: 30–50%, mostly within 2 years.
14. UGN-101 (Jelmyto): FDA 2020, low-grade UTUC; stricture ≈44%.
15. Commonest BCG upper-tract complication: renal granulomas.
16. Upper-tract CIS: 6-week BCG induction, preferably via nephrostomy.
17. Antegrade instillation pressure column ≤25 cm H_2O.
18. Neoadjuvant chemotherapy favoured when post-RNU CrCl expected <50.
19. First ureteroscopic tumour treatment: Goodman 1984; antegrade: Tomera 1982.
20. First laparoscopic nephroureterectomy: Clayman 1991.
21. Renal descensus gains up to 8–10 cm, more on the left.
22. Upper-tract CIS diagnosed by persistently positive selective cytology with normal imaging and URS.
23. Bipolar resection avoids hypo-osmotic irrigant absorption in percutaneous resection.
24. Ureteroscopic resection never goes beyond lamina propria.
25. Ileal ureter: Foley ≥1 week; cystogram + nephrostogram before tube removal.

141 Contemporary Open Surgery of the Kidney

Reading time ≈ 54 min (slow pace)

1 · ROLE & PREOPERATIVE WORK-UP

- Open surgery = **exposure + early great-vessel control**: locally advanced renal cell carcinoma (RCC) with inferior vena cava (IVC) thrombus, cytoreduction, complex tumours, hostile abdomen. Minimally invasive → *Ch 142*; RCC staging → *Ch 137*.
- **Renal function**: urinalysis, serum creatinine (SCr), estimated glomerular filtration rate (eGFR) by the **Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) 2021 race-free** equation, more accurate than Modification of Diet in Renal Disease (MDRD). It decides partial (PN) vs radical nephrectomy (RN), counselling and dialysis planning. Both equations rest on creatinine, so muscle mass and diet distort them. Cystatin C is made at a constant rate by all nucleated cells and detects mild GFR loss that creatinine misses.
- GFR **<60** or significant proteinuria → perioperative nephrology consult; poorly controlled diabetes or hypertension needs optimising even when GFR ≥60.
- Advanced RCC: blood count, metabolic panel (Memorial Sloan Kettering [MSKCC] and International Metastatic RCC Database Consortium [IMDC] risk models), liver tests + coagulation (**Stauffer syndrome** — paraneoplastic hepatic dysfunction), cardiopulmonary assessment, because positioning, blood loss and fluid shifts can decompensate heart and lungs.
- Imaging maps kidney, vessels and neighbours (colon, pleura, pancreas, spleen, liver): computed tomography (CT)/magnetic resonance imaging (MRI); contrast generally safe if GFR **>30**. Suspected thrombus → MR or CT venography, **≤1 month old** (rapid propagation). Venography shows cranial and caudal extent and suggests caval wall invasion, which decide liver mobilisation, level of caval control and need for graft or resection.
- Prophylaxis: cefazolin (or clindamycin) ≤1 h pre-incision; urine culture only if infection suspected; no routine bowel prep; stress-ulcer prophylaxis (proton pump inhibitor or sucralfate) if prolonged intensive care expected; pneumatic compression + heparin 5000 IU subcutaneously; stop smoking 4–6 wk; spirometry.
- **Venous thromboembolism (VTE)**: risk raised by immobility, tissue injury and malignant hypercoagulability. Mechanical prophylaxis for all; drugs for raised risk once bleeding risk falls.
- **Renal artery embolisation (RAE)**: rationale — less blood flow and blood loss, oedematous planes, shrinkage of tumour or arterialised thrombus, and allows vein ligation before artery when bulky hilar tumour or nodes hide the artery. Authors no longer use it routinely. Postinfarction syndrome (pain, fever ≤72 h); may *increase* blood loss (oedema); harder dissection if surgery delayed >48 h. If used, embolise the evening before or ideally the morning of surgery, and confirm inflow is occluded at operation (may be incomplete).
- **Instruments**: table-mounted retractors (Bookwalter, Omni-Tract, Thompson) are stable but complex; Balfour is quick but less stable. Vascular clamps (bulldog, Satinsky, pedicle clamps) have atraumatic jaws — never Kocher clamps on vessels. Monopolar for vessels <2 mm (spray for organ surfaces); bipolar sealers for vessels; argon beam for liver tears.

2 · SURGICAL ANATOMY & APPROACHES

- At risk: **right** — liver, duodenum, colon; **left** — spleen, pancreatic tail, colon, superior mesenteric artery (SMA).
- **Right renal artery**: posterior-inferior to renal vein, behind IVC; reachable in the **interaortocaval space** by retracting the left renal vein (LRV) upward.
- **SMA**: anterior aorta **1–2 cm above LRV**, veers left → mistaken for left renal artery.
- Right gonadal/adrenal veins → IVC; left → LRV.
- Anterior exposure = colon off Gerota's via **white line of Toldt**; right adds **Kocher** (no cautery — duodenal injury devastating); left divides splenorenal ligament.

- **Cattell-Braasch**: Toldt → caecum → Kocher → mesenteric root to Treitz; midgut pivots on SMA, exposing aorta/IVC. **Infracolic approach** (trauma, bulky hilum): incise below Treitz → aorta + LRV. Partial **Mattox** (left tumours): divide splenic diaphragmatic attachments so spleen and pancreas rotate medially together.

Incision	Best for	Limitations
Midline	Trauma; advanced tumour/IVC thrombus (authors' choice — early aortic access); bilateral	Bowel mobilisation
Anterior subcostal	Unilateral transperitoneal	Muscle division
Chevron	Renovascular; RN + thrombectomy; both pedicles	Extensive muscle division
Subcostal flank (below 12th rib)	Lower pole, pelviureteric junction, proximal ureter	Poor hilar access; not for PN
Supracostal (11th/12th)	RN/PN; 8-cm mini-flank supra-11th	Pleura
Transcostal (11th, rib resected)	Retroperitoneum at old rib tip	Rib division
Thoracoabdominal, dorsal lumbotomy	Largely historical (morbidity); thoracoabdominal still an option for level III–IV thrombus	

- Choice of approach depends on operation, imaging anatomy, prior surgery, habitus, kyphoscoliosis and lung disease. Good exposure minimises retraction and therefore retractor injury.
- **Midline**: open linea alba midway between xiphoid and umbilicus (thinnest preperitoneal fat). Benefits: no muscle division, less pain, both kidneys and great vessels accessible.
- **Supine**: lumbar roll/flexion; excess hyperlordosis → back pain (avoid in spinal stenosis); aortocaval compression; pad elbows.
- **Flank**: 12th rib tip over kidney bar, axillary roll; ↓ venous return and lung function — poor for cardiopulmonary disease, severe kyphosis; pad against rhabdomyolysis.
- **Subcostal flank**: curve down at midaxillary line to avoid subcostal nerve; two fingers at 12th rib tip sweep peritoneum medially while transversus is split.
- **Supracostal steps**: incise on upper rib border → open lumbodorsal fascia at rib tip (avoids peritoneum and pleura) → mobilise pleura up → intercostals divided close to upper rib border toward spine.
- Higher rib = better upper-pole/hilar exposure but nearer pleura. Spare **subcostal nerve** (between internal oblique and transversus); cut intercostals on **upper border of rib**.

3 · SIMPLE NEPHRECTOMY

- **Within Gerota's fascia** for benign nonfunction (obstruction, infection, stones, reflux, polycystic, dysplasia); functioning kidney rarely (pain, bleeding, renovascular hypertension, infection).
- Stay on, not in, the capsule; include psoas fascia if adherent; aspirate large hydronephrosis; spare adrenal by keeping on the capsule at the upper pole. Benign disease distorts planes (prior instrumentation, chronic infection, aberrant anatomy), so expect adherent, inflamed perinephric fat.
- Benign PN: focal atrophy, duplex pole, infected diverticulum, symptomatic angiomyolipoma.

4 · RADICAL NEPHRECTOMY

- Robson: kidney + Gerota's + adrenal + nodes crus → bifurcation. Now: **spare normal adrenal**, selective lymph node dissection (LND).
- **Indications**: not PN-amenable — nonfunctioning kidney, tumour replacing kidney, nodes, cytoreduction, venous thrombus.
- Staging: abdominal CT/MRI + **chest CT** (plain film misses small/mediastinal disease); bone scan only for symptoms, fracture, suspicious imaging or raised bone alkaline phosphatase; no routine

brain imaging. Look for thrombus, nodes and vascular/collecting-system anomalies.

- Biopsy if it changes plan: patient wants histology; benign result would change strategy (accepting false negatives — a minority of benign biopsies prove RCC); infiltrative mass (urothelial, lymphoma, metastasis), before systemic therapy, bilateral tumours, trials.
- T4 invasion suggested by altered organ enhancement, not contact. Get MRI and multidisciplinary planning if colon, duodenum, liver or spleen may be involved. Liver rarely truly invaded (bilaminar capsule); if so, depth usually <1 cm and a thin crescent is resected by a liver surgeon.
- Liver retraction: leave the falciform intact if not mobilising the liver (a falciform flap can buttress a duodenal repair), but an intact falciform predisposes to traction lacerations; divide the right triangular ligament if more exposure is needed.

ALGORITHM

Right: survey abdomen → Toldt, iliac → hepatic flexure (protect right colic/ileocolic) → Kocher → IVC → loop and palpate renal vein → artery behind vein → **artery then vein** → kidney outside Gerota's.

Difficult hilum: interaortocaval artery ligation → kidney decompresses → flaccid vein checked, double-ligated (0 silk + 3-0 Prolene) → lateral artery now easy to reach, divided. Or: lower pole → ureter and gonadal vein (spare gonadal if safe; large tumours risk avulsion from IVC/LRV) → divide ureter → lift kidney posterior to anterior to show artery.

Left: Toldt splenic flexure → iliac → renocolic ligament (pad pancreas) → aorta leads to LRV; gonadal vein leads up to it → artery above/behind LRV → 0 silk + 3-0 Prolene, confirm occlusion → vein.

KEY POINT **Why artery first?** Tying the vein first leaves the kidney engorged with arterial inflow and collaterals bleed. Arterial ligation shrinks the kidney and softens the vein so it can be checked for thrombus. A persisting pulse means an accessory artery, incomplete control, or that the SMA was tied.

- Resect Gerota's fascia entire: best chance of cure, no tumour spill.
- **En bloc pedicle** (hilar nodes prevent separating artery and vein): dissect pedicle to 2–3 cm → 3 clamps (first proximal, extending beyond all structures), divide between 2nd and 3rd → tie ×2 with 0 silk + suture ligature (silk slips); if stumps not separately visible, running 3-0 Prolene mattress below the clamp; arteriovenous fistula (AVF) theoretical, very rare.
- **Lumbar veins**: silk ties; **no clips on IVC side** (they block emergency caval clamping). Retracted lumbar: sponge-stick psoas compression → fulgurate/suture.
- **Loss of pedicle control**: tell anaesthetist, resuscitate, finger/sponge compression, 2 suckers, **never clamp blindly**. Artery: compress aorta above, clamp, 2-layer repair. IVC hole: finger → Allis → suture.
- Prolene (double-armed, tapered): IVC/aorta 3-0/4-0; renal vessels 4-0/5-0; ⅜-circle BB needle on arteries (less likely to crack calcified plaque).
- **Adrenalectomy** only for upper-pole tumour with lost plane, adrenal mass, diffuse involvement. Metastasis **<5%**; CT/MRI miss 20–25%, so use clinical indicators. Contralateral metastasis on follow-up is as common as ipsilateral → prophylactic removal risks adrenal insufficiency without benefit.
- **LND**: nodes positive ≈5%. European Organisation for Research and Treatment of Cancer (EORTC) randomised trial (772 patients): no overall survival (OS) benefit, but many small low-stage tumours enrolled. LND is prognostic, may reduce retroperitoneal recurrence. **Do if nodes enlarged** (short axis **≥1.5 cm ≈ 50% positive**); consider in high risk (cytoreduction, >10 cm, grade ≥3, sarcomatoid, thrombus, T4).
- Templates: right — paracaval, precaval, retrocaval, interaortocaval, crus → bifurcation; left — para/retroaortic, LRV → bifurcation. Right: tie gonadal at IVC, ligate lumbars, roll IVC. Left: preserve SMA and coeliac trunk; IMA may be divided. **Clip lymphatic trunks above renal vein**: chyle from cisterna chyli otherwise causes chylous ascites.

- **Closure**: saline in wound + big breath; bubbles = pleural tear → suture over 14-Fr catheter, withdraw on held breath; **chest X-ray in all**, even if not suspected. Fascia in two layers (transversus + internal oblique, then external oblique).

5 · IVC TUMOUR THROMBECTOMY

- RCC = commonest cause; **5–10%** of RCC. Untreated → most die within 1 yr, so surgery (± systemic therapy) is the backbone, with RN + LND.
- Other causes: Wilms tumour, adrenocortical carcinoma, urothelial carcinoma, sarcoma, lymphoma. Thrombus marks aggressive RCC: nodes, metastases and perirenal fat invasion are common.
- **Tumour thrombus** (contains tumour) vs **bland thrombus** (clot below obstruction) — managed differently. MR venography is best for extent, wall invasion and bland thrombus, as it is independent of contrast timing.
- Preop: screen for pulmonary embolism (PE)/deep vein thrombosis; anticoagulate if venous thromboembolism, bland thrombus or IVC occlusion. **No suprarenal filters**: risk of renal/hepatic vein thrombosis and embolisation, and thrombus grows onto the device and invades the IVC, enlarging the reconstruction. Embolise rarely: one-third of thrombi have their own arterial supply, but infarction seldom shrinks them usefully and causes pain. Consider only for unexcisable hepatic-vein thrombus (Budd–Chiari) or a bleeding kidney, within 1 day of surgery.
- **Transoesophageal echo (TEE)** for level II–IV. Vascular/liver surgeon II–III; **cardiothoracic for III–IV**, planned, with cardiac anaesthetist for TEE. TEE (after induction) defines the upper extent, fragmentation and haemodynamics.

Mayo level	Upper extent	Management
0	Renal vein	RN (commonest)
I	IVC **≤2 cm** above ostium	Milk back, Satinsky partial clamp, ostial cavotomy
II	>2 cm, below hepatic veins	Full IVC control; infrahepatic cavotomy
III	At/above major hepatic veins, below diaphragm	Liver mobilisation ± Pringle; occlusive → suprahepatic clamp; nonocclusive → venovenous bypass (VVB)
IV	Above diaphragm	Sternotomy, cardiopulmonary bypass (CPB) + deep hypothermic circulatory arrest (DHCA), atriotomy

Bland group	Location	Action
A	None	None
B	Iliac veins	Infrarenal filter
C	Infrarenal IVC, separate	Interrupt (clip/staple); don't extract — organised clot fails to clear, vessel injured
D	Mixed with tumour	Resect infrarenal IVC; keep lumbars as collaterals

ALGORITHM

Level I (right): usually partially occlusive, nonadherent. Friable or embolised thrombus → control IVC before handling tumour. Midline/subcostal → minimal handling of renal vein and IVC → right artery ligated interaortocavally → loop LRV, suprarenal and infrarenal IVC → **Rummel tourniquets** (18-Fr red rubber tube on vessel loop) → pinch IVC above thrombus, milk thrombus toward ostium → Satinsky around ostium (confirm thrombus inside jaws) → circumferential ostial cavotomy, keeping ≥half IVC width → extract with traction on renal vein → tie gauze over vein stump.

Level I–II: early renal artery ligation → tourniquets **infrarenal IVC → contralateral renal vein → suprarenal IVC** → cavotomy → extract → heparinised saline flush, inspect wall (biopsy/resect suspicious intima) → 4-0 Prolene → release **caudal clamp first**, flush 5–10 mL on held breath → cranial clamp.

- **Left tumour (level II)**: harder — IVC reached from the right, both colons mobilised (midline or chevron). Left side first: ligate left renal artery near aorta and confirm with palpation or Doppler (persisting pulse = accessory artery or incomplete control) → tie dilated, friable adrenal, lumbar and gonadal branches of LRV → mobilise kidney. Then right side: Kocher, expose IVC to bifurcation, **split-and-roll** IVC with lumbar ligation. Excise LRV ostium, extend onto IVC with Potts scissors, extract with Penfield dissector, remove en bloc with kidney. During long clamping, periodically release contralateral renal vein and lower clamps. Authors don't clamp right renal artery (no venous collaterals); intermittently release right renal vein clamp. No routine heparin — bleeding, not clotting, is the problem.
- **Level III**: bypass on standby; transplant-style liver mobilisation. **Major hepatic veins never divided**; accessory (caudate) veins tied → 2–3 cm more IVC. **Budd–Chiari = contraindication**. Decision between abdominal-only extraction and combined thoracoabdominal approach with bypass is made only intraoperatively, after artery ligation, liver mobilisation and IVC assessment.
- Liver mobilisation: divide ligamentum teres, falciform, coronary and triangular ligaments; rotate right lobe medially.
- **Pringle** needed if clamping above hepatic veins (else liver engorges and bleeds): ≤60 min, ideally **≤20 min** (liver ischaemia, portal thrombosis, splenic congestion and rupture). Pringle + clamps above and below liver = total hepatic vascular occlusion. Not needed if clamping below the major hepatic veins with accessory veins tied.
- **Level III sequence**: clamp infrarenal IVC → contralateral renal vein → porta hepatis → suprahepatic IVC → cavotomy from ostium upward → extract; clear adherent thrombus with Penfield dissector; use a Fogarty or 20-Fr Foley as embolectomy catheter for unreachable thrombus; unscrapable wall invasion → partial/complete IVC resection. To shorten Pringle: after extraction move clamp below the liver, release suprahepatic clamp, then Pringle. Tack hepatic ligaments back to prevent liver torsion.
- **Clamp trial** 2–5 min: preload ↓60%, cardiac output (CO) ↓40%. **CO ↓>50% or mean arterial pressure (MAP) ↓>30%** → bypass (preferred) or supracoeliac aortic clamp. Proceed if tolerated and extraction <30 min. Occlusive thrombus tolerates clamping (azygos collaterals); free-floating does not.
- Bleeding and hypotension cause incomplete tumour removal, which is universally fatal, so never compromise IVC control to save time.
- Also bypass for cardiac/hepatic dysfunction, poor contralateral kidney, portal hypertension, uncontrolled bleeding.
- **Level III–IV combined approach** (level III not removable from below; nearly all level IV): midline laparotomy + median sternotomy preferred (alternatives: chevron + sternotomy, thoracoabdominal). Call cardiothoracic surgeon after releasing caudate lobe but before rotating liver → sternotomy → bypass → cavotomy ± right atriotomy → close atrium and IVC → off bypass, chest tubes.

	VVB	**CPB + DHCA**
Access	Femoral → jugular, percutaneous	Sternotomy; 16–18 °C, drain 95% blood
Heparin	Partial	Full; coagulopathy
Drawback	Lumbar/intercostal back-bleeding	Cardiac, neurological
Use	Least invasive; not with atrial or iliac bland thrombus, Budd–Chiari	Level IV; bloodless field; ≤60 min (<40 better)

- **VVB**: cannula tips far from thrombus, as dislodgement causes massive PE and tumour cells can be recirculated.
- **Why DHCA**: usable when thrombus cannot be milked below an intrapericardial clamp; no aortic or portal clamping or extensive vein ligation (but every cut vessel must be controlled before coming off bypass); bloodless view of IVC and hepatic veins; less embolism; longer working time.

- **Reconstruction**: narrowing to 50% → primary closure; more → **patch** (bovine pericardium, smooth side in); width = normal diameter × fraction resected × 3 (Leibovich–Bower); 5-0 Prolene, small regular bites, edges not inverted (rough surface is thrombogenic). Circumferential → **polytetrafluoroethylene (PTFE) 16–20 mm** graft (larger diameter = less thrombosis): wrap in omentum; low-dose heparin then lifelong aspirin 81 mg.
- Suprarenal IVC ligation: right tumour — OK if LRV tied caval to its tributaries (collateral drainage). Left tumour — right kidney has no collaterals → needs alternative drainage (autotransplant or saphenous graft to splenic, portal or inferior mesenteric vein).

> **KEY POINT "Shortcuts" that become "longcuts":** opening IVC before all inflow controlled; side-biting a level I; clips on IVC side; incomplete arterial control; forceful milking (embolus, tear); clamping before volume loading or skipping test clamp.

6 · PARTIAL NEPHRECTOMY

- **Treatment of choice for T1** even with normal function; size-independent if safe. Old imperative indications: solitary kidney, bilateral tumours, renal failure risk.
- **Relative contraindications**: <20% nephron mass left; pedicle encasement, collecting-system invasion, venous thrombus, cT4, cN1.
- **Hyperfiltration**: fewer nephrons each filter more → focal segmental glomerulosclerosis, proteinuria, chronic kidney disease (CKD); rare unless nephron mass ↓>80%.
- Multifocality: clear cell ≈2%, papillary ≈10% → inspect whole kidney. Hereditary: observe to **3 cm**, except **HLRCC** (hereditary leiomyomatosis and RCC) → early wide excision; use intraoperative ultrasound (IOUS). Preserve perirenal fat and Gerota's, see whole surface, resect all visible tumours; reapproximate fat and fascia so later surgery is easier.

Question	Evidence
Warm ischaemia	Aim **<30 min**; within limits, preserved parenchyma dominates
Cold vs none	Randomised controlled trial (RCT): no difference
Mannitol	RCT: no benefit
On- vs off-clamp	2 RCTs: no difference
3D models	RCT: less ischaemia, blood loss, stay

> **KEY POINT** PN trades nephrons lost to the margin against nephrons injured by ischaemia. Within accepted limits, the **volume of parenchyma preserved** matters more than clamp minutes, so ischaemia is individualised; hypothermia and mannitol are optional (role in solitary kidney/CKD unclear).

- Always **control hilum first**, even if not clamping, so a clamp can go on at once. Anticipate retroaortic vein, precaval artery, early branching, accessory vessels. Options: manual compression, Kaufmann parenchymal clamp, selective branch, whole pedicle; early unclamping after excision or first layer; off-clamp. Leaving the vein open has no proven benefit.
- Mobilise at least enough to expose capsule around the tumour for capsular closure; full mobilisation for multiple tumours.
- IOUS by the surgeon after defatting, before clamping: endophytic/central tumours — borders, depth, relation to vessels and collecting system; exophytic tumours follow the visible contour.
- **Enucleation**: pseudocapsule plane, minimal normal tissue. **Resection**: 5–10 mm rim, usually clamped. Enucleation steps: score cortex circumferentially → follow pseudocapsule bluntly/sharply → ligate small intrarenal vessels then divide → inspect margins; frozen section of deep margin if needed. Resection transects more small vessels → meticulous haemostasis.
- **Renorrhaphy**: inner — 4-0 absorbable to vessels, running closure of collecting system (no permanent sutures = stone nidus); outer — capsular sutures over bolster, tension along suture (upward pull tears). Avoid catching neighbouring vessels (delayed haematuria). Unclamp → inspect

→ compress or add sutures until dry → close fat and Gerota's (prevents bowel adhesion) → closed suction drain if collecting system entered.

7 · COMPLICATIONS & MANAGEMENT

Injury	Management
SMA	Supplies small bowel to proximal transverse colon. Mistaken for left renal artery, traction in Cattell-Braasch (midgut pivots on it), retractor blade. Potentially lethal → reverse at once; vascular surgeon, primary repair or graft
Inferior mesenteric artery/vein (IMA/IMV)	Ligation safe if marginal artery of Drummond patent (SMA collaterals).
Superior mesenteric vein (SMV)	Ligation → bowel oedema, ischaemia, splanchnic hypertension with systemic hypotension. Never ligate unless no option; clip tributaries, 6-0 Prolene venorrhaphy; transection → reanastomosis or autologous vein; leave abdomen open (compartment syndrome)
Liver/spleen	Retraction tears. Argon beam or spray cautery, fibrin glue, oxidised cellulose; liver laceration — absorbable suture with pledgets; serious splenic injury → splenorrhaphy/splenectomy
Duodenum	Kocher or Cattell-Braasch. Haematoma: observe; if narrowing lumen, incise serosa and muscularis (not mucosa), drain, close 3-0 silk; tear: debride, mucosa 4-0 absorbable running + seromuscular 3-0 silk, omentum, drain, nasogastric tube; don't resect "nonviable-looking" bowel
Pancreas	Glue + drain (amylase); ductal → general surgeon; late: pain, ileus, raised amylase, left renal-bed collection → percutaneous drain, nil by mouth, antibiotics if infected, octreotide
Pleura	Purse-string over 14-Fr catheter, evacuate air on positive-pressure breath; large → chest tube. Late dyspnoea/chest pain → X-ray, oxygen, chest tube

- **Thrombectomy**: air embolism to right heart/pulmonary arteries (release caudal clamp first, let ≈5 mL blood escape); PE (minimal handling; pulmonary arteriotomy if collapse); bleeding despite control = untied lumbar; **no cell saver** (spreads tumour); liver enzymes peak 2–3 days; post-CPB cardiac events (ischaemia, arrhythmias, heart failure) and neurological events (deficits, seizures, delirium in elderly). Uncontrolled bleeding off bypass → supracoeliac aortic clamp or CPB.
- **Urine leak** (collecting-system entry; large, endophytic, hilar tumours; most appear within 72 h): close big defects primarily; remove drain by 48 h if low. Output >50 mL/24 h → drain creatinine > serum = urine; CT urography only if symptomatic. Drain → radiological drain → if urinoma persists >48 h or symptoms >24 h, retrograde double-J stent + Foley (low-pressure pyelogram). Most resolve 4–6 wk.
- **Bleeding/pseudoaneurysm** (disrupted artery communicating with a contained haematoma, from excision, suturing or thermal injury; risk ↑ with complex tumours and postoperative anticoagulation; flank pain, haematuria, shock if ruptured): stable → bed rest, serial haemoglobin; significant → **angiography** (CT angiography misses pseudoaneurysm, can't treat) + **selective embolisation**; refractory → re-explore.
- **AVF** (high-flow artery-to-vein communication): thrill or pulsatile mass, bruit, haematuria, heart failure, renal ischaemia from steal. Diagnose with Doppler, CT or MR angiography. Observe → coil embolisation or stent → surgery for complex/symptomatic.
- **Renal insufficiency**: usually mild, transient; risk with solitary kidney, large resection, long ischaemia; also exclude obstruction, drug toxicity, vascular thrombosis. Supportive care; some need dialysis.

8 · MCQ ONE-LINERS

1. CKD-EPI 2021 is race-free, preferred over MDRD.
2. Thrombus imaging must be ≤1 month old.
3. Postinfarction syndrome lasts up to 72 h.
4. Subcostal flank: poor hilar access, not for PN.
5. Turner-Warwick popularised supracostal incision. MCQ

6. Right renal artery lies behind renal vein and IVC.
7. SMA arises 1–2 cm above LRV.
8. Left gonadal and adrenal veins drain into LRV.
9. Robson (1969) defined radical nephrectomy.
0. Adrenal metastasis <5%; imaging misses 20–25%.
1. Node short axis ≥1.5 cm ≈ 50% positive.
2. EORTC LND trial: no OS benefit.
3. Venous thrombus in 5–10% RCC.
4. Level I: ≤2 cm above ostium.
5. Level III defined by major, not accessory, hepatic veins.
6. Budd–Chiari contraindicates thrombectomy.
7. Pringle ≤60 min, ideally ≤20 min.
8. Abort clamp trial: CO ↓>50% or MAP ↓>30%.
9. DHCA core 16–18 °C.
0. Patch width = diameter × fraction resected × 3.
1. IVC graft: PTFE 16–20 mm; lifelong aspirin.
22. Bland group D → infrarenal IVC resection.
23. Multifocality: papillary ≈10%, clear cell ≈2%.
24. Hereditary RCC: operate at 3 cm, except HLRCC.
25. Hyperfiltration if nephron mass ↓>80%.
26. Resection margin 5–10 mm.
27. Hypothermia, mannitol, off-clamp: no RCT benefit.
28. Urine leak: drain creatinine > serum.
29. Pseudoaneurysm: angiography over CT angiography.
30. SMV: never ligate.
31. No cell saver in thrombectomy.
32. Supracostal intercostal incision hugs the upper rib border (spares bundle).
33. Accessory hepatic veins drain caudate lobe; safely ligated.
34. Pringle = clamping portal vein, hepatic artery, bile duct.
35. Lymphatics above renal vein clipped to prevent chylous ascites.

142 Laparoscopic and Robotic Surgery of the Kidney

Reading time ≈ 46 min (slow pace)

1 · PRINCIPLES & PATIENT SELECTION

- First laparoscopic nephrectomy: **Clayman, 1991** MCQ. Minimally invasive surgery (MIS) vs open: less pain and blood loss, shorter convalescence, better cosmesis, **equivalent long-term outcomes**. Indications same as open.
- **General anaesthesia mandatory.** CO2 pneumoperitoneum impairs ventilation, venous return and cardiac output. High risk: chronic obstructive pulmonary disease (COPD), intracranial pathology, cardiac disease.
- Severe COPD cannot compensate for CO2 absorption (hypercarbia): lower insufflation pressure → helium → convert to open. AirSeal (valveless trocar, continuous low-pressure high-flow insufflation) reduces CO2 consumption and may improve tolerance.
- Robot: needs a dedicated theatre; **docking/undocking delays emergency access**; console surgeon may miss the patient slipping → close anaesthetist–assistant–surgeon communication.
- Obesity: body mass index (BMI) is a poor predictor of difficulty because it does not separate fat from muscle or visceral from subcutaneous fat; **visceral adipose tissue** is the better index. Elderly: benefits apply irrespective of age.
- History and examination anticipate intraoperative problems; body habitus dictates positioning. Prior surgery alters trocar sites and TP vs RP choice; it is **not a contraindication**. Former absolute contraindications are now relative — selection matters most during the learning curve.

> **KEY POINT** The patient must tolerate the **pneumoperitoneum**, not the renal operation — so cardiopulmonary reserve, not age or BMI, decides fitness.

2 · ACCESS & PORT PLACEMENT

Options (laparoscopic or robotic): transperitoneal (TP, most used), retroperitoneal (RP), hand-assisted laparoscopy (HAL), laparoendoscopic single-site surgery (LESS), natural orifice transluminal endoscopic surgery (NOTES).

	Transperitoneal	**Retroperitoneal**
Space / landmarks	Large; familiar	Limited, poor triangulation; less familiar; lens smudging
Bowel	Colon reflection; adhesiolysis	Peritoneum not entered
Hilum	After colon reflection	Direct, esp. posterior aspect
Preferred for	Most surgery; suturing; anterior tumours	Posterior tumours, pyeloplasty, cysts, biopsy; dense adhesions, peritoneal dialysis, pregnancy, morbid obesity

- **TP positioning:** catheter, compression stockings → modified flank, operative side up 30–45° on gel rolls; pad pressure points, tape securely and tilt the table both ways to test stability; prep the whole abdomen for conversion.
- **Laparoscopic TP ports:** Veress or Hasson (open) → 3–5 trocars: 10 mm umbilical camera; 5/10 mm midline ~2 cm below xiphoid; 12 mm anterior axillary line (AAL) at umbilicus for staplers. Short patient → 12 mm port to lower midline; obese → shift all lateral. Accessory: low midline 10/12 mm (colon retraction), 5 mm subxiphoid (liver retraction).
- **Robotic TP:** 3D view, wristed instruments, tremor elimination, motion scaling. Arm spacing **8 cm (Si) / 6 cm (Xi)**; 12 mm low-midline assistant port for sutures, clamps, staplers, suction (5 mm if no stapler needed, e.g. pyeloplasty).
- Whatever the template, ports form a **broad-based triangle**: camera at apex, working ports at base, target at the far point.

- **RP:** full flank, table flexed, axillary roll; incision in posterior axillary line midway between 12th rib tip and iliac crest → split flank muscles, divide lumbodorsal fascia → finger dissection (feel smooth psoas and lower pole in Gerota's) or balloon dilator (glove fingers, 600–800 mL) → blunt-tip cuffed trocar prevents gas leak. Yellow retroperitoneal fat confirms the plane; peritoneum is swept medially.
- **Robotic RP:** 8 mm camera in midaxillary line 2 cm above iliac crest, arms in posterior and anterior axillary lines; robot docked over the head.

> **KEY POINT** RP pitfalls: balloon inflated within muscle → flank hernia; too anterior → peritoneal/colon injury; too posterior → quadratus/psoas bleeding. Bail-out: open the peritoneum and continue TP.

- **HAL:** gives tactile feedback and manual retraction; useful for novices, donor nephrectomy and scarred kidneys (xanthogranulomatous pyelonephritis, XGP). Drawbacks: forearm paraesthesia, more wound infection/incisional hernia; hand-port metastases → bag the specimen.
- **HAL set-up:** hand-port incision first → pneumoperitoneum → other trocars under vision. Non-dominant hand goes in; incision not too large (gas leak); camera placed relative to the edge of the hand device. Irrigate copiously before closure.
- **HAL emergency role:** extend a port site, insert a hand and compress the bleeding vessel with a finger while open conversion is prepared — keep it in the armamentarium.
- **LESS / NOTES / single-port (SP) robot:** fewer incisions buy cosmesis and less pain at the cost of triangulation. Operative and oncological outcomes unchanged. LESS: longer operation but less analgesia; needs articulating/flexible instruments, often an extra port. da Vinci SP: one 27-mm guide (three instruments + flexible camera), 10–25 cm between cannula and target; SP partial nephrectomy has longer warm ischaemia.

3 · NEPHRECTOMY — OPERATIVE STEPS

ALGORITHM
Reflect colon → find ureter/gonadal vessels on psoas → lift kidney, expose hilum → artery then vein → upper pole (adrenal spared or taken) → ureter and lateral attachments → bag → extract

- **Left colon:** incise white line of Toldt from below the lower pole to above the upper pole; divide lienocolic ligament. Keep the **lateral renal attachments** intact for now — otherwise the kidney rolls over and hides the hilum. The avascular plane lies between bright-yellow mesenteric fat and pale-yellow Gerota fat. Close mesocolic defects (internal hernia).
- Paddle retractor via an extra low midline trocar holds colon, pancreas and spleen.
- **Right:** liver retractor; cold sharp dissection near the duodenum (avoids thermal injury); **Kocher manoeuvre** exposes the inferior vena cava (IVC).
- **Ureter:** follow psoas medially; gonadal vessels are met first, ureter lies just posterior (peristalsis confirms). Do not divide it early — it is the handle for lifting the kidney. Sweep tissue behind ureter and lower pole to bare the psoas, staying above psoas fascia to protect cutaneous nerves (thigh numbness). Right gonadal vein can tear at its IVC insertion.
- **Hilum:** lift kidney anterolaterally to put hilum on stretch; artery usually posterior to vein — map on preoperative imaging. **Artery first, then vein** with an endoscopic gastrointestinal anastomosis (GIA) stapler covering the whole vessel. Left renal vein is stapled lateral to the adrenal vein. **Never leave clips in the staple line** — misfire.
- Grasper under ureter and lower pole, never into parenchyma. Bleeding along the staple line → reinforcing clips. Clips and staplers equally safe; polymer clips 2 mm apart, a few mm from the cut end. **En bloc hilar stapling** is safe (no arteriovenous fistula found) and saves ~45 min.
- **Upper pole:** adrenal-sparing — open Gerota's above the hilum and peel fat off the upper pole. Then clip and divide ureter → lateral attachments → rotate kidney over liver/spleen → divide uppermost attachments under vision. Extreme fibrosis → subcapsular nephrectomy once vessels are controlled. Avoid the diaphragm.

- **Extraction:** intact in a bag via extended port or Pfannenstiel, or morcellation in a **LapSac** (impermeable even after morcellation) under vision, with change of gloves/instruments. Morcellation keeps histological type and grade but loses size and capsular/renal-vein staging. Rules: purpose-built sac, adequate draping; no deep forceps passes (bowel), keep pneumoperitoneum and view of the sac.
- **Post-op:** diet and activity as tolerated; catheter out once ambulant; no heavy lifting 4–6 weeks if extraction incision.

4 · SIMPLE VS RADICAL NEPHRECTOMY

	Simple	Radical
Gerota's fascia and fat	May be left	Removed intact with specimen
Pathology	Benign, often inflammatory	Renal cell carcinoma (RCC)
Difficulty / conversion	Can exceed radical; higher	Planes preserved; lower

- **Simple nephrectomy indications:** chronic pyelonephritis (incl. XGP), obstructive/reflux nephropathy, tuberculosis, multicystic dysplastic kidney, renovascular or post-transplant hypertension, acquired cystic disease on dialysis, nephrosclerosis, symptomatic autosomal dominant polycystic kidney disease (ADPKD).
- Inflammation obliterates planes, so simple nephrectomy can be harder than radical. Predictors: prior pyonephrosis, nephrostomy tube; stone disease has the most complications. Rescue: hand assistance or subcapsular nephrectomy.
- **Radical nephrectomy (RN):** same steps with Gerota's intact. En bloc adrenalectomy → ligate renal vein medial to adrenal vein. RP left: a lumbar vein usually needs division.
- Results: MIS equals open oncologically (5/10 yr) with less morbidity. RP vs TP randomised trials: the only consistent RP advantage is **shorter operative time**.
- **Large tumours (>10 cm):** feasible; problems are reduced space, distorted landmarks (disorientation), and hilum hidden by collaterals or the kidney falling over it → constant reference to imaging, intraoperative ultrasound, extra trocars or hand port for distraction. Conversion higher laparoscopic than robotic.
- **Lymph node dissection (LND):** staging value only; no overall survival benefit. American Urological Association: remove clinically positive nodes. Template: right paracaval/retrocaval/interaortocaval, left para-/pre-aortic (interaortocaval only if suspicious); crus to bifurcation. Clip lymphatics to prevent lymphocele.
- **Venous/IVC thrombus** (experts only): early arterial ligation reduces collateral flow and may retract the thrombus → milk thrombus back → staple beyond, or Satinsky clamp on an IVC cuff. Long thrombus: sequential clamping infrarenal IVC → contralateral renal vein → suprarenal IVC; level III needs liver mobilisation. Intraoperative ultrasound defines thrombus extent. Preoperative angioembolisation helps robotic (not open) surgery by decompressing venous collaterals. → *Ch 141*
- **Cytoreductive nephrectomy** (metastatic RCC): laparoscopy gives quicker convalescence, less blood loss, no more complications → earlier start of systemic therapy.

> **KEY POINT** Access changes morbidity and recovery, never oncology: early arterial control, en bloc removal and intact bagged extraction carry over from open surgery unchanged.

5 · PARTIAL NEPHRECTOMY

- Partial nephrectomy (PN) is the default where feasible: preserved function, equivalent oncology.
- **RENAL nephrometry score** — Radius, Exophytic/endophytic, Nearness to collecting system, Anterior/posterior, Location vs polar lines; predicts complications, warm ischaemia time (WIT), conversion to RN.
- Clinical T1b/T2: PN gives better estimated glomerular filtration rate (eGFR) and less chronic kidney disease (CKD) with equal oncology. **Caveat: tumours >5 cm upstaged to pT3a recur**

more after PN. For large tumours the argument for PN is renal function, not oncology.

- **Central** tumour: abuts renal sinus fat/collecting system. **Hilar**: touches renal artery/vein → consider enucleation; use superficial renorrhaphy bites over a bolster to avoid major vessels.
- Multifocal tumours: nephron-sparing (contralateral recurrence risk); selective segmental clamping guided by 3D reconstruction. **Solitary kidney:** robotic PN gives short WIT and rare conversion to RN.
- **Heminephrectomy** (≥30% parenchyma): deeper resection, large intraparenchymal vessels clipped/sutured, deliberate collecting-system entry and repair.
- Central/hilar tumours: review imaging + intraoperative ultrasound; 3D CT reconstruction maps vessels; all reconstruction must fit the ischaemia time.
- **Approach choice** depends mainly on tumour location (anterior → TP, posterior → RP) and experience; also size, number of tumours and renal arteries, visceral fat, prior surgery. RP series look better partly because simpler tumours are chosen.

ALGORITHM
Intraoperative ultrasound (size, depth) → open Gerota's away from lesion → score capsule → clamp hilum (Satinsky en bloc or bulldogs) → cold-scissor excision → close vessels/collecting system → renorrhaphy → unclamp → drain

- **Trifecta:** negative margin + no major complication within 30 days + **WIT ≤25 min**.
- **Enucleation** follows the pseudocapsule: less blood and function loss, more positive margins, same survival. Use in familial RCC, multifocal disease, solitary kidney, severe CKD.
- **Renorrhaphy:** suturing is most reliable; sliding-clip technique (Lapra-Ty) keeps tension without knots; barbed sutures shorten WIT. **Single-layer** closure gives better function and fewer pseudoaneurysms. Sealants (Floseal, Tisseel, Coseal) widely used: no randomised benefit; sutureless closure possible in selected cases. Large vessels in the bed are sutured or clipped individually.
- **Collecting system** entered in **~60%** (large, endophytic tumours). Repair with 2-0/3-0 polyglactin; **always leave a drain** (urinoma). A routine ureteric catheter does not prevent fistula. Check integrity with intravenous indigo carmine or retrograde instillation; argon beam can destroy the repair suture.
- **Ischaemia:** historical warm limit **30 min**. **Preserved parenchymal volume predicts function better than WIT.** Consider cold ischaemia (ice slush, cold perfusion) for solitary kidney, CKD or long expected clamp time.

Hilar strategy	Best for	Verdict
Artery-only vs artery + vein	Any	Equivalent
Early unclamping	Any	Possibly fewer pseudoaneurysms
Selective / super-selective (near-infrared fluorescence, duplex)	Mapped segmental supply	Randomised: no 6-month benefit
Off-clamp (zero ischaemia)	Small, exophytic	Better early eGFR; no 2-yr difference; more blood loss
Parenchymal compression	Peripheral	Feasible, safe
Hypothermia	Solitary kidney, long WIT	Benefit unproven

KEY POINT Why no clamping strategy has beaten standard clamping on long-term function: what is kept matters more than how long it is clamped. Precise excision saves nephrons; minutes of ischaemia matter less.

- Robotic vs laparoscopic PN: fewer conversions, shorter WIT. Robotic vs open: less blood loss, fewer complications.

- **Positive margins (0–7%)** raise recurrence only in high-risk tumours (pT2–3, grade 3–4) → **surveillance, not reoperation**.
- **Port-site recurrence ~0.1%** — due to spillage and technique, not pneumoperitoneum or gas; prevent with impermeable bag.
- Local recurrence after RN (~2%): resection + systemic therapy. Salvage after PN or ablation is hard because planes are lost (no pseudocapsule, fibrosis) → often RN. After failed cryoablation/high-intensity focused ultrasound (HIFU): perinephric fibrosis → collecting-system injury, bleeding.

6 · OTHER RENAL PROCEDURES

Procedure	Indication	Key points
Cyst decortication	Pain, infection, obstruction, suspected malignancy	Percutaneous aspiration first — proves the cyst causes pain. Dissect wall, aspirate, excise, biopsy, fulgurate base; collecting-system entry → close + drain. ADPKD: decortication, bilateral nephrectomy, renal denervation
Laparoscopic renal biopsy	Failed percutaneous biopsy, bleeding diathesis, solitary kidney, morbid obesity	RP, two ports, lower-pole cortex; more glomeruli
Nephropexy	Descent >5 cm / 2 vertebrae supine → erect, with pain, reduced Doppler flow or obstruction	Usually right, slim young women; 20% asymptomatic. 2-0 polypropylene capsule to psoas/quadratus fascia; avoid iliohypogastric and genitofemoral nerves
Calyceal diverticulectomy	Symptomatic; endoscopy failed	Laparoscopy sutures the neck closed and obliterates the cavity with fat; endoscopy instead widens the neck
Nephrolysis	Chyluria, loin pain–haematuria	Lateralise by cystoscopy; hilar lymphatic ligation ± omental wrap
Ablation	≤3 cm, comorbid, solitary, hereditary RCC	Percutaneous now preferred. Cryo: two freeze–thaw cycles, iceball 0.5–1 cm beyond tumour on real-time ultrasound. Radiofrequency ablation (RFA): 60–100°C (<105°C), 1 cm margin, impedance-guided; more retreatment

7 · COMPLICATIONS & MANAGEMENT

- Overall **~15%**, mostly postoperative; PN > RN > simple. Learning curve ~20 laparoscopic nephrectomies. Robotic PN halves open conversion vs laparoscopic.
- Patient risks: obesity, comorbidity, prior surgery; age is not a major risk. Conversions cluster in infective disease.
- Prior surgery: place Veress/first trocar away from scars, use Hasson access, or go retroperitoneal.
- **Vascular injury is the commonest complication**; major injury <1/1000 but fatal in 8–17%. Occurs at **access** (Veress, first trocar) or hilar dissection. Device failure: staplers > non-locking titanium clips > locking clips; stapler errors are mostly operator (stapling over clips, tissue too thick, early stop).
- **Left RN pitfall:** superior mesenteric artery (SMA) mistaken for renal artery → bowel necrosis; needs open reconstruction.
- Hilar injury causes: dividing structures before identification, sharp hilar dissection, stapler not covering the vessel.
- Venous ooze: gauze pressure for minutes; visible hole → grasper → clip or suture. Brisk arterial bleed: hand through a lower midline incision to compress. **Never clip or suture blindly.**

ALGORITHM
Major bleed: locate → temporary control (grasper/pressure) → inform anaesthetist (blood, open set) → clear field, raise pneumoperitoneum or add trocar → definitive clip/suture → else convert and call vascular surgeon

- End of case: **lower insufflation pressure** to unmask venous bleeding — commonest sites are the dissection bed, adrenal, mesentery, gonadal vessels, ureteric stump.
- Post-op haemorrhage (hypotension, tachycardia, falling haematocrit) → computed tomography (CT); arterial source → angioembolisation. **Haematuria ~day 12 after PN** → pseudoaneurysm/arteriovenous fistula → selective embolisation (endophytic, posterior tumours at risk).
- **Urine leak** after PN (0.8–10%; risk: large or hilar tumour, collecting-system repair, long WIT): percutaneous drainage → if persistent or distal obstruction, ureteric stent + catheter, or nephrostomy.
- **Unrecognised bowel injury** (<1%, mortality ~3%): thermal energy causes about half and presents later; 32% occur at access. Classic picture: **pain at a single trocar site**, distension, diarrhoea, low-grade fever, **normal or low white count**, bowel sounds present. First test: CT with oral contrast. Stable → laparoscopic repair; unstable/gross spillage → laparotomy; controlled fistula → bowel rest and parenteral nutrition. Close left mesenteric defects without compromising colonic supply. Recognised superficial thermal injury → imbricate; devitalised segment → resect.
- **Solid organs:** spleen (left traction) → haemostatic agents/argon → splenorrhaphy → splenectomy; pancreas → suture ± drain or distal pancreatectomy — injury is insidious, so inspect at the end. Liver → haemostatics/argon; gallbladder → cholecystectomy.
- **Diaphragm** (upper pole): sudden rise in airway pressure → purse-string, suction tube withdrawn on full inspiration, chest X-ray; drain large pneumothorax.
- † CO_2 gas embolism: sudden hypotension, desaturation, fall in end-tidal CO_2 → stop insufflation, left-side-down head-down, aspirate via central line.
- **Neuromuscular** (2.7%): rhabdomyolysis with long operations in full flank — buttock/thigh pain, raised creatine kinase, myoglobinuria → hydration, correct acidosis. Prevent with shorter operations and partial rather than full flank. Axillary roll prevents brachial plexus injury.
- **Thromboembolism:** lower than open; post-discharge risk with preoperative weight loss and chronic steroids; low risk → compression devices, high risk → heparin. Also port-site hernia, ileus.

> **KEY POINT** MIS complications are the open ones in altered form and timing: the lethal ones — access vessel injury, thermal bowel injury — happen out of view and present late or atypically.

8 · LAPAROSCOPIC LIVE DONOR NEPHRECTOMY †

All facts in this section are outside the source notes (†).

- **Principle:** the donor is a healthy person, so safety comes first and the donor keeps the better kidney. Graft quality depends on long vessels, a well-vascularised ureter and minimal warm ischaemia.
- **Work-up:** computed tomography (CT) angiography for arteries, veins and collecting system; split function (renogram) if the kidneys differ in size; donor eligibility checks (glomerular filtration rate, blood pressure, body mass index, diabetes, stones, malignancy).
- **Side:** the **left kidney is preferred** because its renal vein is longer. Choose the right if the left has multiple arteries or the right kidney is the poorer one.
- **Key steps (left):** reflect the colon → identify the gonadal vein and ureter → keep the "gonadal–ureteric packet" together to preserve the ureteric blood supply → divide the gonadal, adrenal and lumbar tributaries of the renal vein → dissect the artery to its aortic origin → keep the patient well filled (fluids ± mannitol) to maintain diuresis → prepare the extraction incision (usually Pfannenstiel) → divide the ureter low, then artery, then vein → extract quickly to keep warm ischaemia to a few minutes.

- **Vascular control:** use vascular staplers or transfixion. **Do not rely on locking polymer clips on the donor renal artery**; clip slippage has caused donor deaths.
- **Complications:** bleeding, ileus, chylous ascites, wound problems; donor mortality about 0.03%. Long-term: small absolute increase in end-stage renal disease risk, and hypertension; donors need lifelong annual blood pressure and renal function checks.

9 · MCQ ONE-LINERS

1. First laparoscopic nephrectomy: Clayman 1991; first laparoscopic PN: Winfield 1993.
2. Better predictor of difficulty than BMI: visceral adipose tissue.
3. Robot arm spacing: 8 cm Si, 6 cm Xi.
4. RP vs TP radical nephrectomy: consistent RP advantage is shorter operative time.
5. En bloc hilar stapling: no arteriovenous fistula; saves ~45 min.
6. Trifecta WIT ≤25 min; historical warm ischaemia limit 30 min.
7. Collecting system entered in ~60% of laparoscopic PN.
8. Positive margins 0–7% of MIS PN; matter only in high-risk tumours.
9. Port-site recurrence ~0.1% of laparoscopic cancer operations.
0. >5 cm tumour upstaged to pT3a: more recurrence after PN than RN.
1. Cryo iceball 0.5–1 cm beyond tumour; two freeze–thaw cycles.
2. RFA 60–100°C; >105°C causes vaporisation; no real-time imaging.
13. Device failure: staplers > non-locking clips > locking clips.
14. Bowel injury mortality ~3%; thermal causes about half.
15. Unrecognised pneumothorax after laparoscopic nephrectomy ~0.3%.
16. Neuromuscular injury in 2.7% of laparoscopic urological surgery.
17. Learning curve ~20 laparoscopic nephrectomies.
18. Commonest MIS complication: vascular injury.
19. Robotic PN morbidity plateaus at 18–20 hospital cases/year.
20. Retroperitoneal access of choice: peritoneal dialysis, dense adhesions.
21. Ureter lies just posterior to gonadal vessels; peristalsis identifies it.
22. Haematuria ~day 12 after PN: pseudoaneurysm → selective embolisation.
23. Balloon inflated within flank muscle → postoperative flank hernia.
24. Nephroptosis: >5 cm or 2 vertebral bodies; asymptomatic in 20% of women.

143 Focal Therapy for Renal Tumours

Reading time ≈ 28 min (slow pace)

1 · RATIONALE & INDICATIONS

- Small renal cell carcinoma (RCC) rarely kills; chronic kidney disease (CKD) and comorbidity do. So treatment minimises competing-risk harm rather than maximising radicality. Small renal mass (SRM) biology/biopsy → *Ch 137*
- **Partial nephrectomy (PN) = gold standard** for clinical T1a (cT1a). Thermal ablation (TA) – cryoablation (CA), radiofrequency (RFA), microwave (MWA) – is the alternative.
- **Indications:** high surgical risk; solitary kidney; bilateral tumours; hereditary syndromes, e.g. von Hippel–Lindau (VHL), where repeat treatments are expected.
- No dissection/renorrhaphy → faster recovery, fewer complications; renal function ≈ or better than PN.

KEY POINT **Trap:** TA is never the default for a fit patient.

Guideline	Position on TA
American Urological Association (AUA) 2021	Consider for cT1a **≤3 cm**, esp. comorbid; PN prioritised; percutaneous preferred
European Association of Urology (EAU) 2022	TA not routinely **>3 cm**; CA not **>4 cm**; for frail/comorbid
National Comprehensive Cancer Network (NCCN) 2022	**Only one allowing cT1b** (non-surgical candidates); may need multiple treatments
American Society of Clinical Oncology (ASCO) 2017	Option if complete ablation achievable

2 · SELECTION & WORK-UP

ALGORITHM
Counsel (age, comorbidity, CKD risk, size/location) → biopsy → re-counsel with histology → PN / surveillance / percutaneous ablation

- **AUA: biopsy all**, preferably at a **separate earlier session**: non-diagnostic biopsy can be repeated, fewer benign masses ablated, better-informed consent. Routine post-ablation biopsy adds nothing over modern CT/MRI; biopsy only a new enhancing lesion.
- Urine culture; treat urinary tract infection (UTI) first – colonised urine (ileal conduit) risks abscess.
- Complication predictors: RENAL nephrometry score (low 4–6, moderate 7–9, high 10–12), American Society of Anesthesiologists (ASA) score >3 (~3×), $(MC)^2$ (CA only).
- Day-case: ≤3 cm, low complexity, no anticoagulation, few comorbidities; else 23-h stay.

3 · MECHANISMS

- Heat and cold share two zones: **central** direct necrosis, and **peripheral sublethal** apoptosis, ischaemia and immune response. They differ only in the direction of the temperature gradient.
- **Cryo kill:** freezing – extracellular ice dehydrates cells, then intracellular ice ruptures them; thawing – water floods back → lysis. Vascular injury: endothelial damage → platelet activation → thrombosis → ischaemia. Immunomodulation: necrotic cells release antigens and damage-associated molecular patterns (DAMPs) → dendritic cells → T cells.
- Kidney dies at −19.4 °C, tumour may need −50 °C → target **≤−40 °C**. Iceball edge is 0 °C; <−20 °C only 3 mm inside it → iceball **5–10 mm beyond tumour**. Every cryo rule (margin, multiple probes, double cycle, duration) exists to push the −40 °C isotherm beyond the tumour edge.
- Heat: 40–45 °C sublethal; **≥60 °C** coagulative necrosis; ≥100 °C vaporisation/charring. Lethal: 45 °C for 60 min, 55 °C for 5 min, **70 °C for 1 min**.

- **Heat sink:** adjacent large vessels carry heat/cold away → surviving perivascular tumour and treatment failure. Counter with colder or longer freezes, more probes, or MWA.

4 · CHOICE OF ENERGY

RFA electrode	Feature	Strength
Single monopolar	Uninsulated tip	Tumours ≤2 cm only
12-tine umbrella	Tines act as antennas	Most spherical zone
9-tine starburst	Thermistors in tines	Best reproducibility
Cool-tip	Saline-cooled 17 G electrode	Largest zone

	Cryoablation	RFA	MWA
Physics	Argon expansion cools (**Joule–Thomson**); helium warms	450–1200 kHz current → ionic friction	915 MHz/2.45 GHz → **dielectric hysteresis**
Target	≤−40 °C	≥70 °C; set 90–105 °C	Often >100 °C
Cycles	**Double freeze–thaw** (10 + 6–8 min)	Two, 30-s cool-down	Single dose (e.g. 65 W × 5 min)
Margin	Iceball 5–10 mm	≥5 mm	≥5 mm
Charring limits	No	**Yes**	No
Grounding pad	No	Yes	No
Heat sink	Yes	Yes	**Least**
Zone visible	**Iceball**	**No** – impedance/temp	Gas bubbles on US
Pros	Spares collecting system	Longest follow-up	Fast; fewest complications
Cons	Bleeding, tumour fracture	Heat sink, pad burns	Risk to adjacent organs

> KEY POINT **Why RFA is self-limiting:** charring raises impedance and halts current, so temperature is capped at 105 °C and electrodes are cooled. MWA *radiates* energy, so it works through desiccated tissue and blood, heats actively to 2 cm (RFA: millimetres) and outruns the heat sink.

- Double freeze–thaw: thaw itself kills and a second freeze enlarges necrosis. 10 min is a compromise – 5 min under-treats, 15 min fractures tissue.
- Thaw: passive, or active (helium). Common hybrid – passive between cycles, active at the end so bleeding is addressed sooner.
- RFA: temperature- vs impedance-controlled generators equivalent; multitine electrodes best. Wet RFA (saline infusion) → larger but less controlled zone; bipolar → elliptical zone, not adopted.
- MWA antenna cooling protects the track and allows higher power; RFA cooling prevents charring.
- RFA impedance protocol: start 40–80 W, raise 10 W/min to a maximum of 130–200 W until impedance reaches 200–500 Ω. The RFA zone is not visible, so it is inferred from electrode choice, accurate placement and generator feedback. Independent thermometry probes at the tumour periphery help.
- MWA settings are manufacturer-derived, so know the zone size and shape your system gives for each time–power combination. Zone size rises with time; in pigs, power beyond 45 W added nothing. Usual cT1 settings are 45–65 W for 5–15 min. Drawbacks: harder energy distribution, bulky cables that heat.

5 · TECHNIQUE

- **Percutaneous = standard** (laparoscopic use has fallen sharply since 2010). Laparoscopic is equally effective but more morbid and costly – keep for prior scarring or failed hydrodisplacement.

- **General anaesthesia preferred:** controlled respiration improves probe placement; high-frequency jet ventilation (~120/min, small tidal volumes) reduces renal motion. Conscious sedation only if intubation not tolerated.

ALGORITHM
IV antibiotics → prone/flank on computed tomography (CT) gantry → ultrasound (US), CT or magnetic resonance imaging (MRI)-guided placement → non-contrast CT confirms position → ablate → contrast CT (completeness, haematoma)

- **Hydrodisplacement:** needle into plane between tumour and bowel/ureter/organ → saline, or 5% dextrose (**D5W for RFA**, †saline conducts current) ± contrast; aim e.g. colon >1 cm. Add saline slowly during ablation to stop overheating. Caution in cardiac failure, because the fluid is absorbed.
- **Pyeloperfusion** (central tumours): ureteral catheter irrigated at the **opposite temperature** – cold for RFA/MWA, warm for CA.
- **CA:** multiple probes **1–2 cm apart**; remove only after full thaw, without resistance – early removal fractures tumour and bleeds. Size the iceball on CT, where it is a distinct hypodense zone, or on real-time US. Tumours >4 cm can be ablated, but bleeding rises with size and probe number.
- **RFA:** ≥70 °C in each quadrant during cool-down excludes heat sink; ablate track only past Gerota fascia (spare nerves). Position the electrode for complete coverage plus a ≥5 mm margin. On contrast CT, under-treated areas → reposition and repeat; larger lesions → overlapping ablations.
- **MWA:** antennas tangential, never pointing into collecting system; 1 antenna ≤2.5 cm, 2 for 2.5–3.5, 3 for 3.5–4.5 cm. Cooled antenna tip anchors it. Continuous US shows gas bubbles covering the tumour, then contrast CT should show no enhancement including a ≥5 mm margin. Poor renal function → unenhanced CT plus contrast-enhanced US.

6 · FOLLOW-UP, SUCCESS & SALVAGE

- **Success = complete loss of contrast enhancement** on CT/MRI. The tumour stays in situ, so an imaging margin replaces the pathological margin – hence biopsy first and long surveillance.
- **Incomplete ablation:** enhancement on the *first* scan → repeat ablation. Immediate post-procedure contrast imaging allows same-session retreatment and should be standard.
- **Local recurrence:** initially non-enhancing, *later* enlarges/enhances → biopsy; surveillance, re-ablation or excision.
- **Enlargement after any modality = recurrence until proved otherwise.**
- Appearance: CA shrinks >50% in year 1; RFA minimal shrinkage, **fibrotic halo**; MWA low-attenuation zone that keeps contracting; MRI T1-hypointense, non-enhancing.
- **AUA:** CT/MRI with and without contrast + chest imaging at **6, 12, 24, 36, 48, 60 months**; after 5 years, shared decision. Consider risk-adjusted follow-up for larger or high-grade tumours. US is not routine unless a contrast-enhanced US (CEUS) protocol exists.
- Salvage: re-ablation, PN, radical nephrectomy (RN), or stereotactic radiotherapy if comorbid.

7 · ONCOLOGICAL OUTCOMES VS PN

- No randomised comparison. Local recurrence-free survival (LRFS) worse after TA but **equal once re-ablation counted**; **metastasis-free and cancer-specific survival (MFS, CSS) equal**; overall survival lower (older, comorbid).
- **Size predicts failure:** RFA worse ≥3 cm; T1b worse than T1a. CA for T1b recurs more than PN. NCCN alone allows TA for T1b. No energy superior; guidelines accept all three – choose by institutional preference and tumour.

KEY POINT **Why comparisons are biased:** benign tumours and under-graded biopsies flatter TA; occult pT3a and growth-selected tumours penalise it. Older guidelines restricted TA to non-surgical candidates, so many patients die of other causes before progression is detected. European Organisation for Research and Treatment of Cancer (EORTC) 30904 (PN vs RN, ≤5 cm) had only 12 RCC deaths at 9.3 years – trials need huge cohorts.

8 · COMPLICATIONS & MANAGEMENT

- Major (Clavien ≥3) **≤5%**, fewer than PN; percutaneous < laparoscopic. Laparoscopic complications fall with centre experience. Pain is minimal: modern cryoprobe shafts are insulated, and MWA has shaft cooling and no grounding pads.
- Complications are geographical: they follow the probe path and the thermal field. Prevention is therefore planning (probe count, trajectory, organ displacement, protecting the collecting system) rather than intra-operative rescue.

Complication	Prevention	Management
Haemorrhage – commonest; transfusion 2–3%	Full thaw before CA probe removal; avoid intercostal arteries	Bleeding at placement → start ablating (RFA/MWA coagulate); laparoscopic → haemostatic agents + pressure; persistent → selective embolisation; wall haematoma → serial imaging/haematocrit
Urinoma, fistula, pelviureteric junction (PUJ)/ureteric stricture	Stent + pyeloperfusion; CA for central; tangential antennas	Haematuria → conservative, embolise if significant; leak/obstruction → conservative or stent; large urinoma → percutaneous drain. Rarely delayed cutaneous fistula
Bowel/organ (tumour <1 cm away)	Hydrodisplacement, positioning	Surgical consult; colonephric fistula → stent + low-residue diet; persistent → diversion; peritonitis → explore
Pneumothorax (above 12th rib)	Recognise during procedure (dyspnoea, imaging)	Aspirate at end; larger → blood patch; chest tube rarely
Nerve injury / RFA pad burn	Tumour falls away from wall; pads level on posterior thighs	Neuralgia/neuropraxia self-limiting
Infection <1%, may be lethal; abscess/drainage at 1–6 months	Urine culture, prophylaxis	†Drain + antibiotics

9 · NEWER MODALITIES (INVESTIGATIONAL)

- **High-intensity focused ultrasound (HIFU):** entirely extracorporeal (no seeding, bleeding or leak), thermal + cavitation above 3500 W/cm^2; one transducer treats and monitors; ablate-and-resect studies showed incomplete ablation (breathing, acoustic interference).
- **Histotripsy:** focused ultrasound, mechanical (non-thermal), real-time imaging; no clinical data yet (HOPE4KIDNEY trial).
- **Stereotactic radiotherapy (SBRT/SABR):** no anaesthesia, anticoagulation continued; but no imaging endpoint (little change after treatment), viable attenuated tumour on biopsy, lower-than-expected CSS. Possible role: salvage after TA/PN in comorbid patients.
- **Irreversible electroporation (IRE):** non-thermal; the electric field opens membrane nanopores, reversible or lethal according to voltage; spares matrix and vessels; poor oncological control.

10 · MCQ ONE-LINERS

1. AUA: TA for cT1a ≤3 cm; only NCCN allows cT1b.
2. Argon cools, helium warms (Joule–Thomson).
3. Cryo target ≤−40 °C; iceball 5–10 mm beyond tumour.

4. MWA works by dielectric hysteresis; RFA by ionic friction.
5. >70 °C for 1 min lethal to RCC.
6. Only RFA needs grounding pads and is limited by charring.
7. MWA least susceptible to heat sink.
8. Only CA shows a visible real-time zone.
9. After RFA: fibrotic halo; after CA: >50% shrinkage in year 1.
0. Pyeloperfusion temperature opposes the ablation.
1. CA preferred for central tumours.
2. Commonest complication: haemorrhage; major ≤5%.
3. Lowest complication rate: MWA (network meta-analysis).
4. CSS and MFS equal TA vs PN.
15. Histotripsy is non-thermal; HIFU thermal + cavitation.
16. IRE preserves extracellular matrix and vessels.
17. Normal renal parenchyma dies at −19.4 °C.
18. <−20 °C reached only 3 mm inside iceball edge.
19. RFA 450–1200 kHz; MWA 915 MHz or 2.45 GHz.
20. D5W, not saline, for hydrodisplacement with RFA.
21. Success = complete loss of enhancement on CT/MRI.
22. AUA follow-up imaging: 6, 12, 24, 36, 48, 60 months.
23. Biopsy before ablation, ideally at a separate session.
24. Laparoscopic ablation: prior scarring or failed hydrodisplacement.
25. 12-tine umbrella RFA electrode gives the most spherical zone; cool-tip the largest.
26. HIFU cavitation threshold >3500 W/cm².

144 Treatment of Advanced Renal Cell Carcinoma

Reading time ≈ 36 min (slow pace)

1 · OVERVIEW

- Renal cell carcinoma (RCC) is a family of distinct cancers. About **1/3** present with synchronous metastases; a further **20–40%** of localised tumours later metastasise. Metastatic RCC (mRCC) 5-yr survival ≈ **12%**; 10-yr <5%.
- Three treatment eras: cytokines → single-agent targeted therapy → **immune checkpoint inhibitor (immuno-oncology, IO) combinations**.
- Clear cell RCC (ccRCC) is the main target of all these drugs; non-clear cell RCC (nccRCC) has little prospective evidence.
- Pathology, staging and nomograms for localised disease, and adjuvant therapy (KEYNOTE-564, S-TRAC) → *Ch 137*.

2 · PROGNOSTIC MODELS

- Good prognosis: long interval from diagnosis to metastasis, few metastatic sites. Bad: poor performance status (PS), lymph-node or liver metastases.
- **MSKCC (Memorial Sloan-Kettering Cancer Center, Motzer) model**: cytokine era, 670 patients mostly on interferon-α (IFN-α). **IMDC (International Metastatic RCC Database Consortium, Heng) model**: re-derived in patients on first-line vascular endothelial growth factor (VEGF)-targeted therapy; validated in 849.

Factor	MSKCC (5)	IMDC (6)
Performance status	Karnofsky <80%	Karnofsky <80%
Haemoglobin	< lower limit of normal (LLN)	< LLN
Corrected calcium	**>10 mg/dL**	> upper limit of normal (ULN)
Lactate dehydrogenase (LDH)	**>1.5 × ULN**	—
Time factor	No prior nephrectomy	**<1 yr** diagnosis → systemic therapy
Neutrophils; platelets	—	> ULN (each a factor)
Favourable (0) – median overall survival (OS)	20 mo	43.2 mo
Intermediate (1–2)	10 mo	22.5 mo
Poor (≥3)	4 mo	7.8 mo

KEY POINT Both models score host reserve (PS, anaemia), tumour-driven systemic effects (calcium, LDH, neutrophils, platelets) and pace of disease. Three factors are shared; IMDC swaps LDH and prior nephrectomy for neutrophilia, thrombocytosis and a short interval. Poor risk = ≥3 factors in both. The IMDC risk group now **decides the first-line regimen**.

3 · BIOLOGY AND DRUG TARGETS

- **VHL (von Hippel-Lindau) gene**, chromosome 3p: classic tumour suppressor (second hit needed). Its protein, with elongins B/C and cullin-2 (CUL2), tags **hypoxia-inducible factor (HIF)-α** for ubiquitin-mediated degradation after prolyl hydroxylation in normoxia.
- When VHL is mutated or silenced (promoter hypermethylation) – as in most sporadic ccRCC – HIF-α accumulates even in normoxia and constitutively drives VEGF, platelet-derived growth factor (PDGF), transforming growth factor-α (TGF-α), glucose transporter-1 (GLUT-1) and erythropoietin. This makes the VHL–HIF–VEGF axis the logical drug target.

- Each targeted drug blocks one consequence of VHL loss: the ligand (bevacizumab), its receptors (tyrosine kinase inhibitors, TKIs), HIF translation (mammalian target of rapamycin [mTOR] inhibitors) or HIF-2α itself (belzutifan). Resistance through MET, AXL and fibroblast growth factor receptor (FGFR) explains why later TKIs are broader.
- **Checkpoints**: cytotoxic T-lymphocyte antigen-4 (CTLA-4; a CD28 homologue binding B7) and programmed death-1 (PD-1, on T cells) with its ligands PD-L1/PD-L2 (tumour, macrophages) switch off antitumour T cells. Checkpoint inhibitors release this brake rather than stimulating immunity nonspecifically, so toxicity is autoimmune and benefit shows as OS and durability rather than progression-free survival (PFS).
- VEGF inhibitors are immunomodulatory (sunitinib reduces regulatory T cells; sunitinib/sorafenib suppress tumour-associated macrophages) – the rationale for IO + TKI combinations.
- ccRCC is chemoresistant (response 5.5–6%; multidrug resistance [MDR] inhibitors failed) and hormone-insensitive (medroxyprogesterone ~2%).

Agent	Target / route	Signature toxicity / note
Bevacizumab	Antibody vs VEGF-A ligand; IV	Bleeding, proteinuria, hypertension (HTN)
Sorafenib	VEGFR2, PDGFR-β, raf-1	Rash, hand–foot syndrome (HFS); now rarely used
Sunitinib	VEGFR2, PDGFR-β, c-KIT, FLT3; **50 mg, 4 wk on / 2 off**	Diarrhoea, HFS, fatigue, HTN, thrombocytopenia, **hypothyroidism**
Pazopanib	VEGFR, PDGFR, KIT	**Hepatotoxicity (↑alanine aminotransferase (ALT))**; less fatigue/HFS
Axitinib	VEGFR1–3 only – most selective	Diarrhoea, fatigue, HTN
Tivozanib	VEGFR1–3, c-KIT, PDGFR-β	Third line
Cabozantinib	VEGFR, **MET, AXL**	Fatigue, diarrhoea, palmar–plantar erythrodysaesthesia; frequent dose reduction
Lenvatinib	VEGFR1–3, **FGFR1–4**, PDGFR-α, RET, KIT	Partner of everolimus or pembrolizumab
Temsirolimus	mTOR; **IV weekly**	Mucositis, rash, ↑glucose, ↓phosphate, ↑cholesterol, pneumonitis
Everolimus	mTOR; oral daily	Asthenia, anaemia
Belzutifan	**HIF-2α inhibitor**; oral	After IO and TKI; also VHL-associated RCC
Nivolumab, pembrolizumab	Anti-PD-1; IV	Immune-related: colitis/enteritis, hypophysitis, vitiligo
Avelumab	**Anti-PD-L1**; IV	
Ipilimumab	Anti-CTLA-4; IV	

4 · SURGERY IN METASTATIC DISEASE

Cytoreductive nephrectomy (CN)

- **Rationale**: rare spontaneous regression of metastases after nephrectomy (<1%, often transient); bulky primaries suppress T-cell function; cytokines rarely shrink the primary. Nephrectomy alone does not alter outcome – it only helps as a platform for systemic therapy.
- **Drawback**: perioperative morbidity and rapid progression mean a significant proportion never reach systemic therapy (only ~half in the high-dose interleukin-2 [IL-2] series; <1/4 if ECOG [Eastern Cooperative Oncology Group] PS 2, multiple sites, brain or liver metastases).
- **Cytokine era**: SWOG 8949 and EORTC trials – CN + IFN-α improved OS without changing response rate, so benefit was not via better cytokine response.
- **TKI era**: CARMENA – sunitinib alone noninferior to CN + sunitinib in MSKCC intermediate/poor risk. SURTIME – deferring CN until after sunitinib was safe, OS favoured deferral.

- **CARMENA caveats**: ~45% poor risk (not normally offered CN); ~25% of CN arm off-protocol; 17% of sunitinib arm had later CN; per-protocol noninferiority lost; investigators could exclude low-burden patients; stopped early, underpowered; sunitinib no longer standard. It does **not** show harm in fit, low-burden, favourable patients.
- **Unlikely to benefit**: poor PS, comorbidity, rapidly progressive disease, brain metastases, poor risk. Role in nccRCC unclear. Authors offer CN to **well-selected** ccRCC patients.

ALGORITHM
mRCC → histology + IMDC risk + PS + metastatic burden
Fit, low burden, favourable features, resectable primary → consider upfront CN → systemic therapy
Intermediate/poor risk or high burden → systemic therapy first → deferred CN in responders

KEY POINT CN helps only patients fit enough, and with disease slow enough, to reach effective systemic therapy. When systemic therapy is active, operating first on poor-risk patients mainly delays it – hence selection and deferral.

Metastasectomy

- No randomised trial; retrospective series. Isolated **pulmonary** metastases are most often resected with curative intent; 5-yr survival **35–50%** after complete resection of limited disease.
- Favourable: **complete resection**, solitary lesion, age <60, disease-free interval >1 yr, lung site, size <4 cm, metachronous. Complete resection still helps with multiple lesions.
- Every predictor is also a marker of indolent disease, so the operation selects good biology as much as it treats it; still a reasonable, widely used practice.

Palliative surgery and radiation

- **Palliative nephrectomy**: intractable pain, haematuria, constitutional symptoms, paraneoplastic syndromes (hypercalcaemia, erythrocytosis, thrombocytosis, HTN). Try non-surgical options first: medical control of pain and hypercalcaemia; **angioembolisation for haematuria**. Only ~half of hypercalcaemic patients improve after resection, so purely palliative nephrectomy is infrequent.
- **Palliative metastasectomy**: solitary brain metastasis; lesions in weight-bearing bones or joints; vertebral metastases with impending cord or root compromise – usually combined with **radiation** and/or systemic therapy.

5 · SYSTEMIC THERAPY FOR CCRCC

Cytokines

	IFN-α	High-dose IL-2
Response	10–15%	15–20%
Durable complete response (CR)	<2%	**7–9%**; >60% of CRs stay disease-free
Regimen	Schedule/route immaterial	**600,000–720,000 IU/kg** IV bolus 8-hourly, max 15 doses
Toxicity	Constitutional	**Vascular leak**: hypotension, third-spacing, respiratory and multiorgan failure; early mortality 2–5%
Current role	Not used first-line	Carefully selected fit patients

- High-dose IL-2 survives because it alone produces durable complete remissions. Low-dose regimens gave fewer durable CRs in two randomised trials, so only high dose is recommended. IL-2 toxicity is controlled by careful selection, intensive monitoring, early IV fluids, vasopressors and antibiotics.
- IL-2 combinations (IFN, 5-fluorouracil, tumour-infiltrating lymphocytes) add toxicity, not survival. Best IL-2 candidates: clear cell, good PS, prior nephrectomy; worst: >1 metastatic site, metastasis within 1 yr, liver metastases.

First-line therapy by IMDC risk

IO/TKI or IO/IO combination is preferred by both the National Comprehensive Cancer Network (NCCN) and the European Association of Urology (EAU). Every first-line approval since 2018 is a checkpoint combination tested against sunitinib.

Risk	Preferred / standard	Alternatives (incl. if IO not possible)	Certain circumstances
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Avelumab + axitinib, cabozantinib, ipilimumab + nivolumab, pazopanib, sunitinib	**Active surveillance**, axitinib, high-dose IL-2
Intermediate / poor	Same three **+ ipilimumab + nivolumab**	Cabozantinib, avelumab + axitinib, pazopanib, sunitinib	Axitinib, high-dose IL-2, **temsirolimus**

KEY POINT **Choosing**: IO/TKI gives higher response and PFS with better tolerance; IO/IO (ipilimumab + nivolumab) gives deeper, durable responses (CR 9%) at the cost of autoimmune toxicity – they have never been compared directly. Ipilimumab + nivolumab is restricted to intermediate/poor risk because favourable-risk patients did better on sunitinib in CheckMate 214. Avelumab + axitinib is approved but not NCCN-preferred (no OS benefit). PD-L1 testing does not select patients for any regimen. Single-agent TKI or temsirolimus is the fallback when IO is contraindicated.

Subsequent therapy – switch mechanism

Prior therapy	EAU standard	NCCN recommended (no preferred)
TKI only (IO-naive)	**Nivolumab or cabozantinib** (alternative axitinib)	IO combinations, nivolumab, cabozantinib, lenvatinib + everolimus
Prior IO	Any VEGF-targeted agent not previously used	Axitinib, **belzutifan**, cabozantinib, lenvatinib + everolimus, **tivozanib**

- Whichever axis was used first, the next line attacks the other (or an unused VEGF target). IO re-challenge after IO is only "useful in certain circumstances".
- Cabozantinib works after VEGF failure because MET and AXL are HIF-driven escape routes. Belzutifan: after both PD-1/PD-L1 and VEGF TKI. Tivozanib: after ≥2 prior therapies. Everolimus alone is supplanted, but used with lenvatinib.
- Failed combinations: sunitinib + mTOR inhibitor and nivolumab + sunitinib/pazopanib (too toxic); bevacizumab + mTOR inhibitor (RECORD-2, INTORACT – no gain).

Toxicity and its management

- **Ipilimumab + nivolumab**: grade 3–4 events **46%**; immune-related events needing high-dose glucocorticoids 35%; permanent discontinuation 22%. The chosen dose (nivolumab 3 mg/kg + ipilimumab 1 mg/kg 3-weekly × 4) was the least toxic in CheckMate 016. Nivolumab alone is better tolerated than everolimus (grade 3–4: 19% vs 37%).
- † Immune-related adverse events: hold IO and start corticosteroids for grade ≥2 (prednisolone 1–2 mg/kg for grade 3–4), tapering over ≥4 weeks; grade 4 usually means permanent discontinuation.
- † Endocrinopathies (hypothyroidism, hypophysitis) are managed with hormone replacement; check thyroid function and cortisol if fatigue or hypotension develops.
- **TKIs**: HTN, diarrhoea, HFS, fatigue, hypothyroidism – managed by dose reduction/interruption; few stop permanently. Avoid pazopanib if hepatotoxicity risk.
- **mTOR**: monitor glucose, lipids, phosphate; watch for pneumonitis.

6 · NON-CLEAR CELL RCC

- 15–25% of kidney cancers; no agent of proven efficacy, so ccRCC drugs are used with modest activity. **Offer trial enrolment to all.**

- **Papillary**: sunitinib response 5–10%; erlotinib (EGFR inhibitor) modest. Temsirolimus beat IFN in the non-clear cell subgroup (~20%) of its poor-risk trial. Everolimus (RAPTOR) modest; sunitinib vs everolimus – neither clearly better, trend to sunitinib. Chromophobe: trend to longer PFS on everolimus.
- **Hereditary papillary RCC (HPRC)**: germline activating **MET** mutation (chromosome 7) → bilateral multifocal type 1 papillary. MET is the hepatocyte growth factor (HGF) receptor; mutation makes it constitutively active. Sporadic papillary: somatic MET ~13%, **trisomy 7 (HGF + MET) in >2/3**. Foretinib (MET + VEGFR2): about half of germline MET carriers responded vs few others; selective MET inhibitors (savolitinib, crizotinib) active in MET-driven tumours.
- **Hereditary leiomyomatosis and RCC (HLRCC)**: germline fumarate hydratase (FH) loss → aggressive type 2 papillary. Fumarate causes VHL-independent HIF activation, and Krebs-cycle failure makes cells glycolysis-dependent and sensitive to glucose deprivation – hence **bevacizumab + erlotinib** to restrict glucose delivery (partial response ~2/3).
- **Collecting duct**: resembles urothelial carcinoma → **gemcitabine + carboplatin**. **Sarcomatoid**: gemcitabine-based chemotherapy promising (small series); more PD-1-positive infiltrates.

> **KEY POINT** In nccRCC the hereditary syndrome names the driver, and the driver names the drug: VHL → VEGF/mTOR/HIF-2α; MET → MET inhibitors; FH → glucose restriction.

7 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SWOG 8949 / EORTC	CN + IFN vs IFN	OS 11.1 vs 8.1; 17 vs 7 mo; response same	Basis of CN in cytokine era
CARMENA	CN + sunitinib vs sunitinib; noninferiority; n=450	OS 13.9 vs **18.4** mo, hazard ratio (HR) 0.89	No upfront CN in intermediate/poor risk
SURTIME	Immediate vs deferred CN	OS 15.0 vs **32.4** mo, HR 0.57	Systemic therapy first is safe
Sunitinib vs IFN	Phase III, first line	PFS 11 vs 5 mo; response 31 vs 6%	Comparator of all later trials
COMPARZ	Pazopanib vs sunitinib; noninferiority	PFS HR 1.05; OS similar	Equivalent; differ in toxicity; quality of life and patient preference (PISCES) favoured pazopanib
CALGB 90206 / AVOREN	Bevacizumab + IFN vs IFN	PFS ↑, OS not	Bevacizumab little used
Temsirolimus phase III	Temsirolimus vs IFN vs combination; poor risk, all histologies	OS 10.9 vs 7.3 mo	Only mTOR inhibitor with OS gain
TARGET	Sorafenib vs placebo after cytokines	PFS 5.5 vs 2.8 mo; OS same	Historical
Everolimus phase III	Everolimus vs placebo after sunitinib/sorafenib	PFS 4.0 vs 1.9 mo; OS same	Now supplanted as monotherapy
Lenvatinib ± everolimus	Phase II, after VEGF therapy	Combination PFS longer than everolimus	Approval of lenvatinib + everolimus
AXIS	Axitinib vs sorafenib, second line	PFS 6.7 vs 4.7; OS same	Second-line axitinib
METEOR	Cabozantinib vs everolimus after VEGF therapy	PFS HR 0.58; **OS HR 0.70**	Only TKI with OS gain over active comparator
CABOSUN	Phase II; cabozantinib vs sunitinib; intermediate/poor	PFS 8.2 vs 5.6, HR 0.66	First-line cabozantinib

Trial	Design	Result	Take-home
TIVO-3	Tivozanib vs sorafenib, after 2–3 lines	PFS HR 0.73; OS same	Third-line tivozanib
LITESPARK-005	Belzutifan vs everolimus after IO + TKI	PFS HR 0.75 (medians equal – curves separate late)	HIF-2α inhibition approved
CheckMate 025	Nivolumab vs everolimus after TKI; primary OS	OS 25.0 vs 19.6, **HR 0.73**; PFS same	IO benefit shows as OS
CheckMate 214	Ipilimumab + nivolumab vs sunitinib	Intermediate/poor OS HR 0.63; CR 9 vs 1%	Only IO/IO; not for favourable
KEYNOTE-426	Pembrolizumab + axitinib vs sunitinib	PFS and OS ↑; response 60 vs 40%	All risk groups
JAVELIN Renal 101	Avelumab + axitinib vs sunitinib	PFS ↑; OS not significant	Not NCCN-preferred
CheckMate 9ER	Nivolumab + cabozantinib vs sunitinib	PFS 16.6 vs 8.3, HR 0.51; OS HR 0.70	All risk groups
CLEAR	Pembrolizumab + lenvatinib vs sunitinib	PFS 23.9 vs 9.2, **HR 0.39**; response 71%	Largest PFS/response effect

8 · MCQ ONE-LINERS

1. MSKCC: LDH >1.5 × ULN, calcium >10 mg/dL, no prior nephrectomy.
2. IMDC adds neutrophilia, thrombocytosis; drops LDH.
3. Poor risk = ≥3 factors; IMDC poor-risk median OS 7.8 mo.
4. VHL: chromosome 3p; complexes with elongin B/C and CUL2.
5. HPRC: MET, chromosome 7, type 1 papillary.
6. Sporadic papillary RCC: trisomy 7 in >2/3.
7. HLRCC: fumarate hydratase; type 2 papillary; bevacizumab + erlotinib.
8. Most selective VEGFR TKI: axitinib.
9. Only anti-PD-L1 used in RCC combinations: avelumab.
0. Lenvatinib uniquely inhibits FGFR1–4; cabozantinib inhibits MET and AXL.
1. Sunitinib: 50 mg, 4 weeks on, 2 off; causes hypothyroidism.
2. Pazopanib: raised ALT; COMPARZ noninferior to sunitinib.
3. High-dose IL-2: 600,000–720,000 IU/kg 8-hourly, max 15 doses.
4. IL-2 dose-limiting toxicity: vascular leak syndrome.
5. IL-2 durable CR 7–9%; only agent approved on phase II response.
6. Temsirolimus: IV weekly; only mTOR inhibitor with OS benefit.
17. Temsirolimus is listed for intermediate/poor risk only; active surveillance for favourable only.
18. CheckMate 025 primary endpoint OS, HR 0.73.
19. CheckMate 214: CR 9% vs 1%.
20. CLEAR PFS HR 0.39 – largest among IO combinations.
21. JAVELIN Renal 101: only combination without OS benefit.
22. First-line regimens limited to intermediate/poor risk: ipilimumab + nivolumab, cabozantinib, temsirolimus.
23. PD-L1 status does not select first-line therapy.
24. Belzutifan targets HIF-2α; approved after IO and TKI, and for VHL disease.
25. Chemotherapy response in ccRCC: 5.5–6%.
26. Collecting duct carcinoma: gemcitabine + carboplatin.
27. Spontaneous regression after nephrectomy: <1%.
28. Metastasectomy 5-yr survival after complete resection: 35–50%.
29. SURTIME: deferred CN OS 32.4 vs 15.0 mo.
30. CARMENA: ~45% MSKCC poor risk; sunitinib alone noninferior.
31. Haematuria from unresectable primary: angioembolisation.
32. Autoimmune events with ipilimumab predicted response.

★ Putting it together: the big treatment choices

Reading time ≈ 10 min (slow pace) · Integrates Ch 136–144 · use these tables as the skeleton of long-essay answers

A · SMALL RENAL MASS (SRM): WHICH TREATMENT?

Four options exist for a localised solid renal mass: active surveillance (AS), thermal ablation (TA), partial nephrectomy (PN) and radical nephrectomy (RN). The choice balances cancer control against kidney function and the patient's fitness. Renal mass biopsy (RMB) is used when its result would change this choice.

	Active surveillance	**Thermal ablation**	**Partial nephrectomy**	**Radical nephrectomy**
Best for	Elderly, frail, life expectancy <5 y; solid <2 cm or mainly cystic	cT1a **≤3 cm**, comorbid; biopsy first	cT1a standard of care; imperative in solitary kidney, bilateral, hereditary tumours or chronic kidney disease (CKD)	Only when all three: high complexity, no CKD/proteinuria, expected estimated glomerular filtration rate (eGFR) **>45**
Key numbers	Growth 0.1–0.3 cm/y; metastasis <2%. Intervene if >3 cm or growth >5 mm/y	Cryoablation: 2 freeze–thaw cycles; radiofrequency (RFA) / microwave (MWA): heat >60–70 °C	"Trifecta": negative margin, no complications, warm ischaemia ≤25 min. Margin width is immaterial	Predicted new GFR ≈ GFR × contralateral split function × 1.25
Weakness	Needs adherence to imaging	More local recurrence (treated by re-ablation); no final pathology	Urine leak, bleeding, pseudoaneurysm	CKD in 65% after RN vs 20% after PN
See	Ch 137	Ch 143	Ch 137, 141, 142	Ch 137, 141, 142

B · RENAL LESION AT A GLANCE

Lesion	**Imaging clue**	**Genetics / immunohistochemistry**	**Default action**
Simple cyst	≤20 Hounsfield units (HU), no enhancement	—	Nothing
Complex cyst	Bosniak v2019 class	—	Size-based: ≤2 cm surveil; >4 cm excise
Angiomyolipoma (AML)	Macroscopic fat ≤−20 HU; fat-poor AML is the trap	TSC2/TSC1; HMB-45, Melan-A positive	>6 cm, aneurysm >5 mm or bleeding → selective embolisation; tuberous sclerosis → everolimus
Oncocytoma	Central scar, segmental enhancement inversion	KIT+, CK7 single cells; Birt–Hogg–Dubé	Biopsy cannot exclude chromophobe RCC → surveil or PN
Clear cell renal cell carcinoma (ccRCC)	Hypervascular, heterogeneous	3p loss / VHL → HIF-2α; CA-IX	PN, RN, TA or AS
Papillary RCC	Hypoenhancing	Trisomy 7, 17; loss of Y; MET	As RCC
Chromophobe RCC	Moderate enhancement	Multiple chromosome losses; diffuse CK7	As RCC; good prognosis

Lesion	Imaging clue	Genetics / immunohistochemistry	Default action
Lymphoma / metastasis	Infiltrative, hypovascular, adenopathy	—	**Biopsy; do not operate**
Upper tract urothelial carcinoma (UTUC)	Filling defect, ~46 HU, positive cytology	Microsatellite instability → Lynch	Risk-stratify → kidney-sparing or nephroureterectomy

C · UTUC: KIDNEY-SPARING MANAGEMENT (KSM) VS RADICAL NEPHROURETERECTOMY (RNU)

	KSM (ureteroscopic, percutaneous, segmental resection)	RNU with bladder cuff
Who	Low risk: ALL of unifocal, <2 cm, low-grade cytology and biopsy, no invasion on computed tomography urography (CTU). Imperative: solitary kidney, CKD, bilateral disease	High risk: ANY of multifocal, >2 cm, high grade, hydronephrosis, previous cystectomy, variant histology
Extras	Ureteroscopy at every surveillance visit; mitomycin gel (UGN-101) for low grade; about 3× more re-interventions	Within 12 weeks; template lymph node dissection; single post-operative dose of intravesical chemotherapy
Systemic	—	Neoadjuvant platinum if post-op creatinine clearance will fall <50; adjuvant platinum (POUT) for ≥pT2 or node-positive

D · OPEN VS MINIMALLY INVASIVE (LAPAROSCOPIC/ROBOTIC) KIDNEY SURGERY

Choose open (Ch 141)	Choose laparoscopic / robotic (Ch 142)
Locally advanced RCC (especially inferior vena cava thrombus needing bypass); cytoreductive setting; anatomically complex tumours; significant previous abdominal surgery. Trauma: midline incision with early vascular control	Standard for RN of organ-confined tumours and most PNs: less pain, blood loss and hospital stay, same cancer outcomes. The retroperitoneal route gives shorter operating time

E · METASTATIC RCC IN FOUR STEPS

ALGORITHM

1. Risk group (IMDC, International Metastatic RCC Database Consortium; 6 factors): Karnofsky <80%, haemoglobin < lower limit of normal, calcium, neutrophils or platelets above upper limit of normal, <1 year from diagnosis to systemic therapy → 0 favourable · 1–2 intermediate · ≥3 poor.

2. First line: immunotherapy (IO) + tyrosine kinase inhibitor (TKI) (pembrolizumab + axitinib, nivolumab + cabozantinib, pembrolizumab + lenvatinib) for any risk; ipilimumab + nivolumab for intermediate/poor risk.

3. Cytoreductive nephrectomy: upfront only if fit, low tumour burden and favourable risk; otherwise systemic therapy first and deferred nephrectomy in responders (CARMENA, SURTIME).

4. Later lines: switch mechanism; belzutifan (HIF-2α inhibitor) after both IO and TKI.